BLUE TEAM / SOC
Wazuh SIEM Lab Project

Document language: English

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1. Project Profile

What has been built up and what practical benefits the project demonstrates

Short description

In a demarcated Hyper-V laboratory environment, a central blue team stack with Wazuh was established. Windows Server 2022, Windows 11 and Ubuntu deliver security-relevant events to Wazuh Manager at CentOS. Kali Linux is used exclusively as an authorized test system. The overall project combines host-based Windows and Linux telemetry, Sysmon, Sigma rules and Suricata network data with reproducible detection, triage and incident response sequences.

Results at a glance

Result	Evidence
Central view of Windows and Linux events	Three active agents; search and analysis in Wazuh Dashboard
Cross-platform FIM	File added, modified and deleted; Content change via syscheck.diff
Windowsity and AD monitoring	Event IDs 4625, 4720, 4728, 4729 and 4726
Own scan detection	Rules 100110 and 100111; correlated level 10 alert; MITRE T1046
Sysmon/PowerShell detection	Sysmon Event ID 1; Wazuh rule 92057; level 12; MITRE T1059.001
SCA and Risk Acceptance	Ubuntu curing with Before/After-Nachweis and justified exception
Incident-oriented editing	Triage, classification, runbooks, cleaning and final checks
Sigma rule development	whoami /priv; Sigma CLI with 0 syntax and condition errors
Suricata network detection	ICMP and TCP-SYN; Rule 100121, Level 8, MITRE T1046

Competences proven

- SIEM basic operation: Check agents, managers, indexers, dashboards and log sources.
- Detection Engineering: Analyze detection gap, develop rules, test correlation and assign MITRE techniques.
- Alert Triage: jointly evaluate agent, user, source, time, rule ID, level and follow-up activity.
- Windows Security Monitoring: Active Directory, PowerShell, Sysmon and Windows Event Logs.
- Linux Security Monitoring: Technically evaluate PAM, FIM and CIS-based SCA results.
- Documentation: Keep track of results, limitations, cleaning and risk acceptance.

2. Architecture and Data Flow

Small, clearly defined laboratory environment with reproducible system roles

Systems and roles

System	IP address	Role	Wazuh
Windows Host	192.168.20.1	Hyper-V, Gateway, NAT	No agent
DC-Test	192.168.20.2	Windows Server 2022, AD DS, DNS, IIS	Agent 001
ham1	192.168.20.5	Windows 11 Domainclient, Sysmon	Agent 002
ubuntu-Desktop	192.168.20.20	Ubuntu Desktop, FIM, PAM, SCA	Agent 003
Kali Linux	192.168.20.7	Authorized tests	No agent
Wazuh Server	192.168.20.8	Manager, indexer, dashboard	Local server

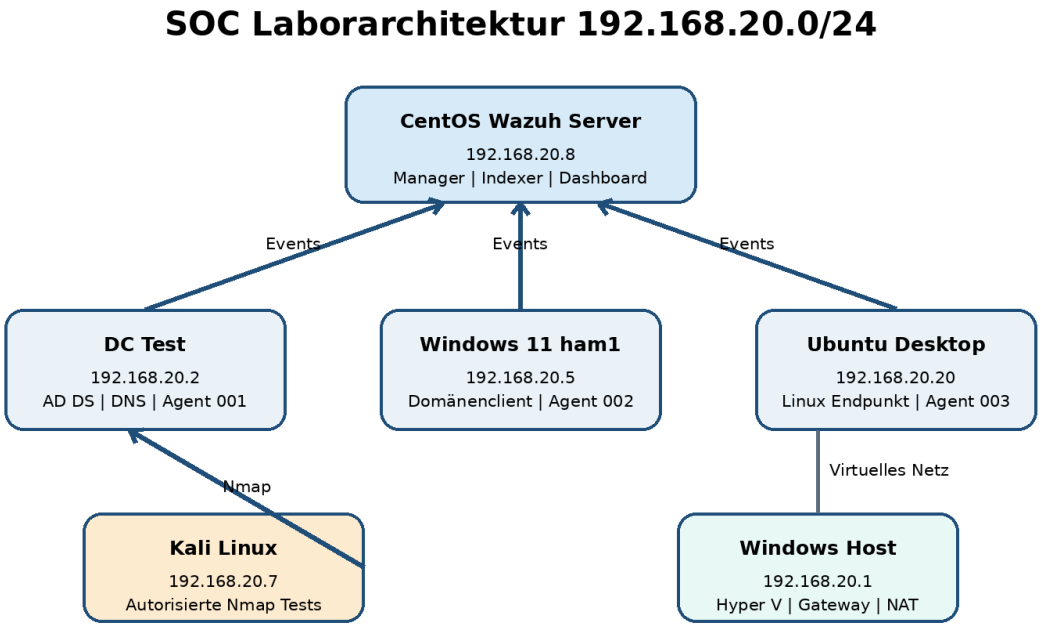


Figure 1: Architecture and data flow of the SOC laboratory environment

Data flow

The endpoints send their events to the Wazuh Manager. There, the data is decoded, evaluated with standard and local rules and stored in the indexer. The dashboard provides alerts for search, filtering and triage. Fixed IP addresses and consistent timestamps facilitate the assignment and correlation.

Laboratory limit tests exclusively in the internal laboratory network; no external systems.

3. Monitoring Platform Architecture

From basic operation to the verified chain of events

Wazuh components

CentOS Wazuh Manager, Indexer and Dashboard. DC-Test, ham1 and ubuntu-Desktop were registered as agents. Data collection was limited to relevant sources to keep the analysis clear.

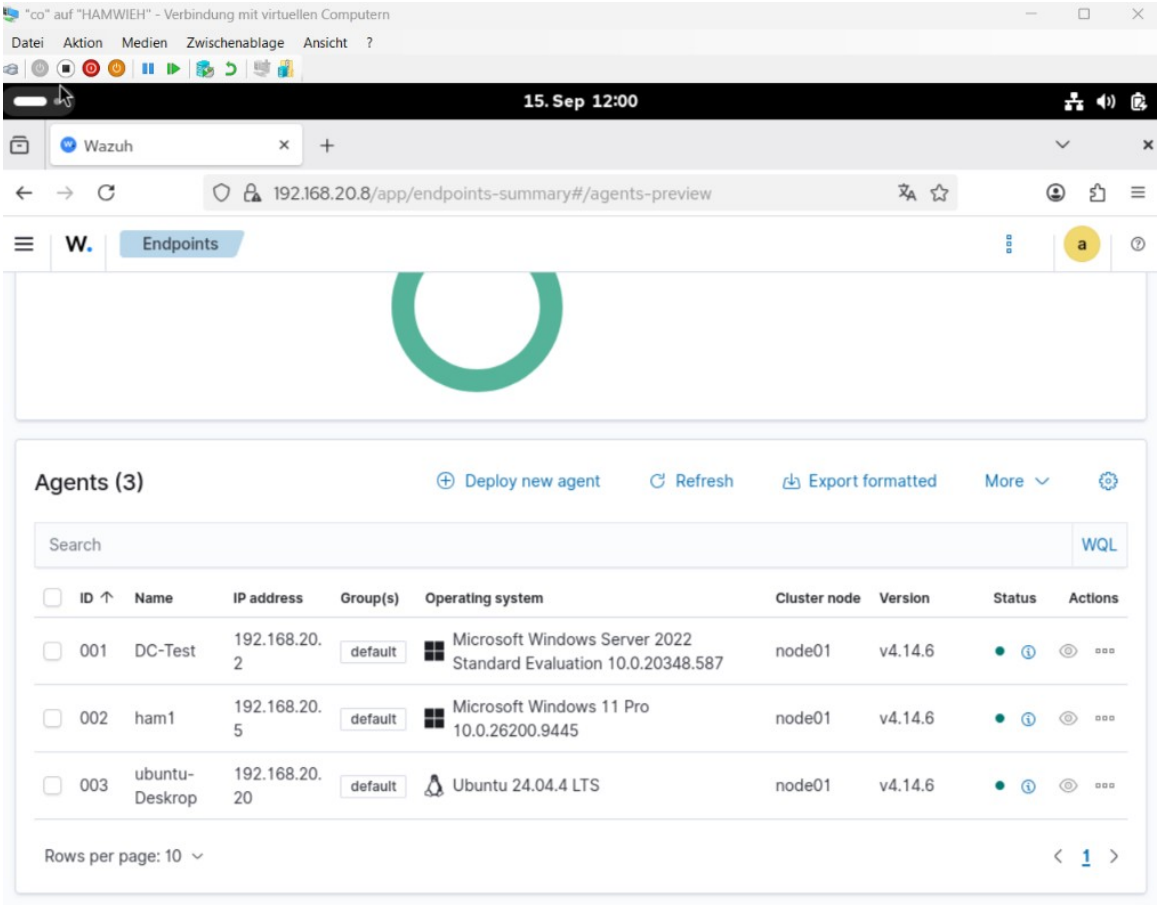


Figure 2: Three active Wazuh Agents with associated endpoints

Quality assurance

- Each use case was triggered in a controlled manner and first checked locally, then in SIEM.
- For each successful alert, at least agent, event or rule ID, level and context were controlled.
- Own rules were syntactically validated before restarting.
- Temporary accounts and privileged changes were removed after the test and rechecked.
- Missing alerts were examined separately after auditing, transport, decoding and regulations.

4. Detection Use Cases

Tested events, rules and analyst evaluation

Use Case	Data source	Detection	Evaluation
Ubuntu FIM	Wazuh FIM	Rules 554, 550, 553; syscheck.diff	Fulfilled
Ubuntu authentication	PAM / auth logs	Rule 5503, Level 5	Fulfilled
Windows Registration	Security Log	Event 4625 / Rule 60122, level 5	Fulfilled
PowerShell Script Block	PowerShell Operational	Event 4104	Fulfilled
AD account life cycle	Security Log	4720, 4728, 4729, 4726	Fulfilled
Privileged Group	Security Log	Rule 60159	Fulfilled
Network Scan	Windows Filtering Platform	100110 + 100111, level 10, T1046	Fulfilled
Ubuntu SCA	Wazuh SCA	Check 35517 Passed; 35509 documented exception	Reviewed
Sysmon PowerShell	Sysmon Event ID 1	Rule 92057, level 12, T1059.001	Fulfilled
Sysmon Privilege Discovery	Sysmon Event ID 1	Custom Rule 100120; whoami /priv	Fulfilled
Suricata ICMP	eve.json / SID 1000001	Wazuh Rule 86601	Fulfilled
Suricata TCP SYN Scan	eve.json / SID 1000002	Rule 100121, Level 8, T1046	Fulfilled

Valuation principle

An event is only a technical record. An alert shows that a rule has been triggered. Only the triage decides whether an incident arises from it. Therefore, the alert level was never only considered, but also source, user, time, authorization and subsequent activity.

5. Endpoint Monitoring

File changes, registrations and PowerShellity

File Integrity Monitoring

FIM was set up for selected paths at Ubuntu and Windows 11. Files were created, modified and deleted in controlled tests. Wazuh showed the actions added, modified and deleted; in addition, the content was understandable via syscheck.diff. The paths were deliberately narrowly selected so that there was no unnecessary event volume.

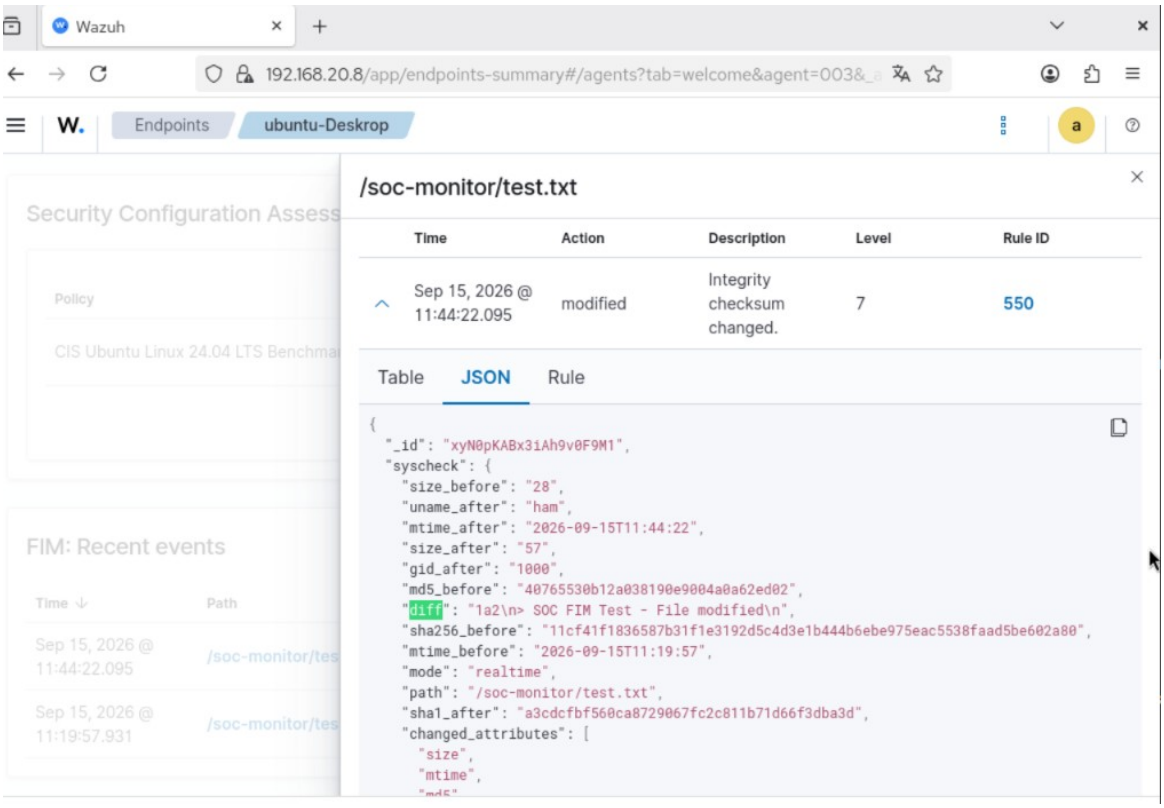


Figure 3: FIM detail view with documented change of content to Ubuntu

Failed Windows application

A failed login attempt was created on ham1. Windows logged event ID 4625. Wazuh took over the data set as rule 60122 at level 5. Users, source address, login type, frequency and successful follow-up logins are relevant for the triage.

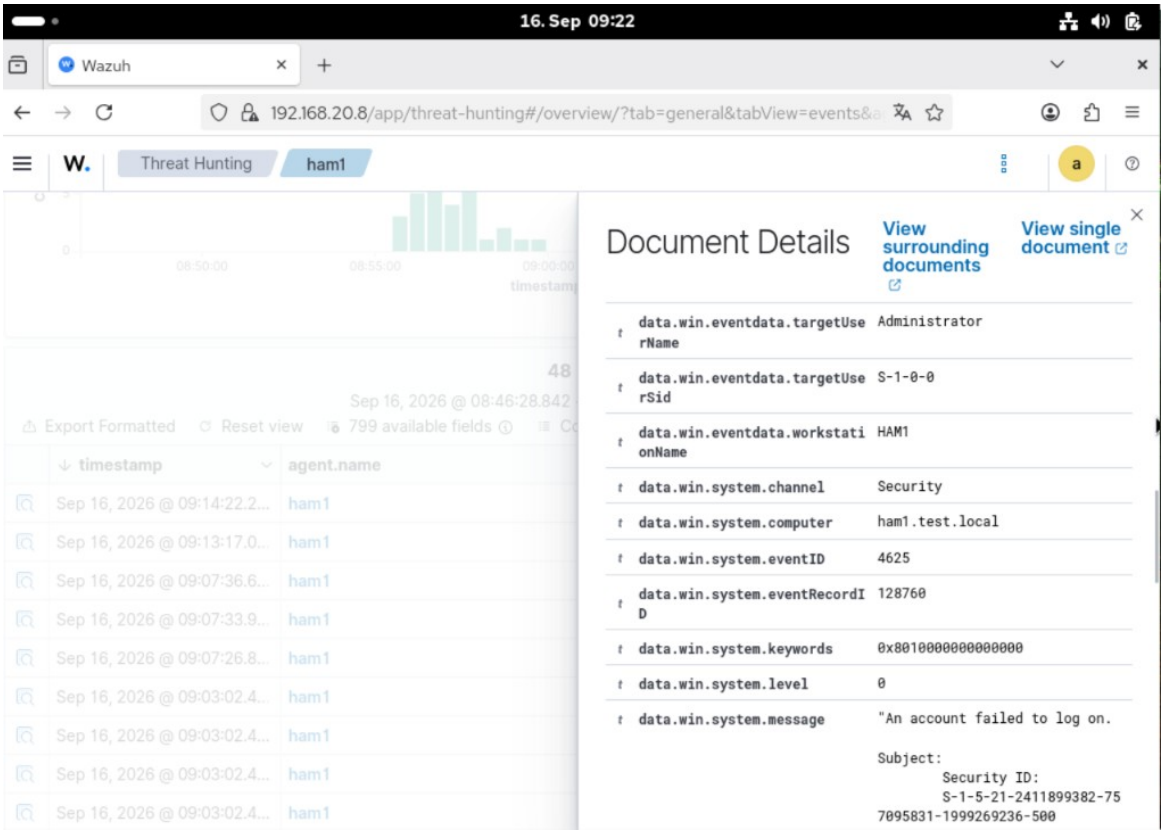


Figure 4: Wazuh evidence of Windows event 4625 at ham1

PowerShell Script Block Logging

A controlled test generated Event ID 4104. PowerShell is also an administration tool and possible attack path; therefore, the Event ID alone is not sufficient for an incident decision. Content, user context, parent process and other events must be evaluated together.

6. Active Directory Monitoring

Recognize and completely clean up privileged changes

A temporary account was created on DC-Test, added to Domain Admins, removed and deleted after that. Events 4720, 4728, 4729 and 4726 represented the life cycle. The privileged group change was highlighted by Rule 60159.

Step	Windows Event	Analyst question
Account created	4720	Who created the account when?
Added to Domain Admins	4728	Was there an approved change?
Removed from Domain Admins	4729	Is the privileged membership completely over?
Account deleted	4726	Are group memberships or other artifacts left behind?

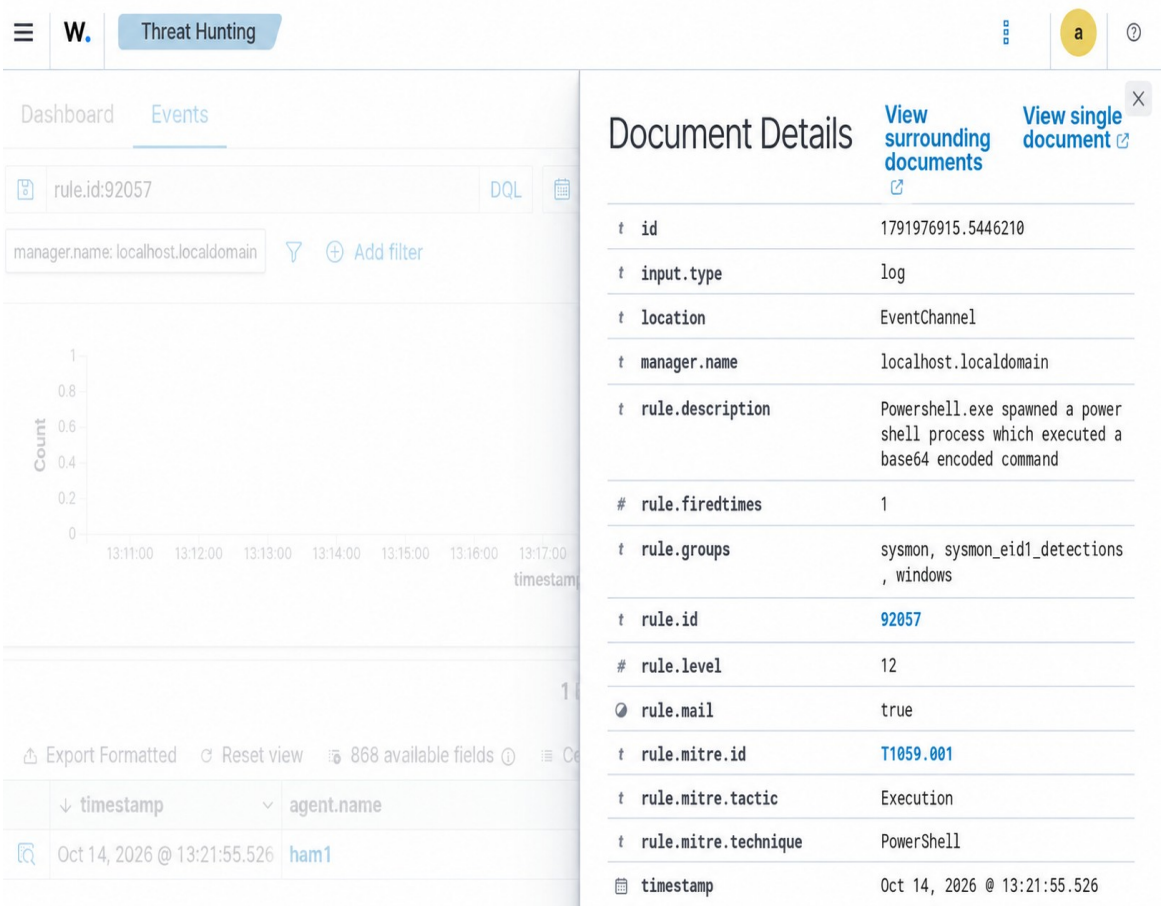


Figure 5: Wazuh rule 60159 for a change in Domain Admins

Final control

After the test, it was checked that the temporary account no longer exists and that no membership in Domain Admins remains. Thus, not only the detection but also the safe cleaning was proven.

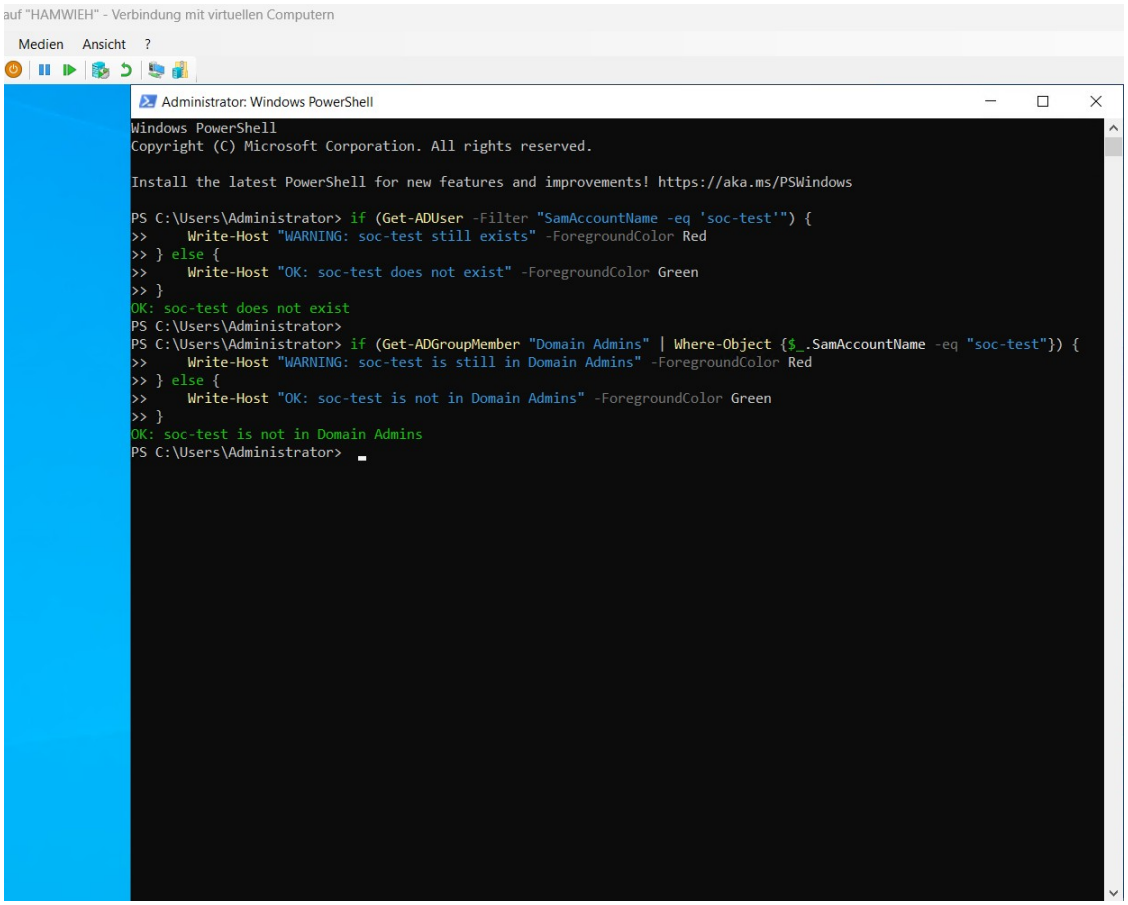


Figure 6: Final control of the remote AD test account

7. Detection Engineering: Network Scan

From raw event to correlated detection

Starting position and detection gap

A limited Nmap test of Kali Linux against DC-Test generated Windows filtering platform events with Event ID 5156. Network activity was visible in the operating system but did not initially result in a matching Wazuh alert. The agent worked; a project-specific rule was missing.

Rule logic

Rule	Function	Result
100110	Recognizes matching 5156 events from the well-known Kali source 192.168.20.7.	Single event, level 5
100111	Corresponds to multiple hits of the basic rule within the defined time window.	Scan alert, level 10
MITRE T1046	Arranges the behavior Network Service Discovery.	Analyst context

The rules were validated with wazuh-analysed -t; the test ended with Exit Code 0. The alert was a true positive, but due to the approved laboratory test no incident.

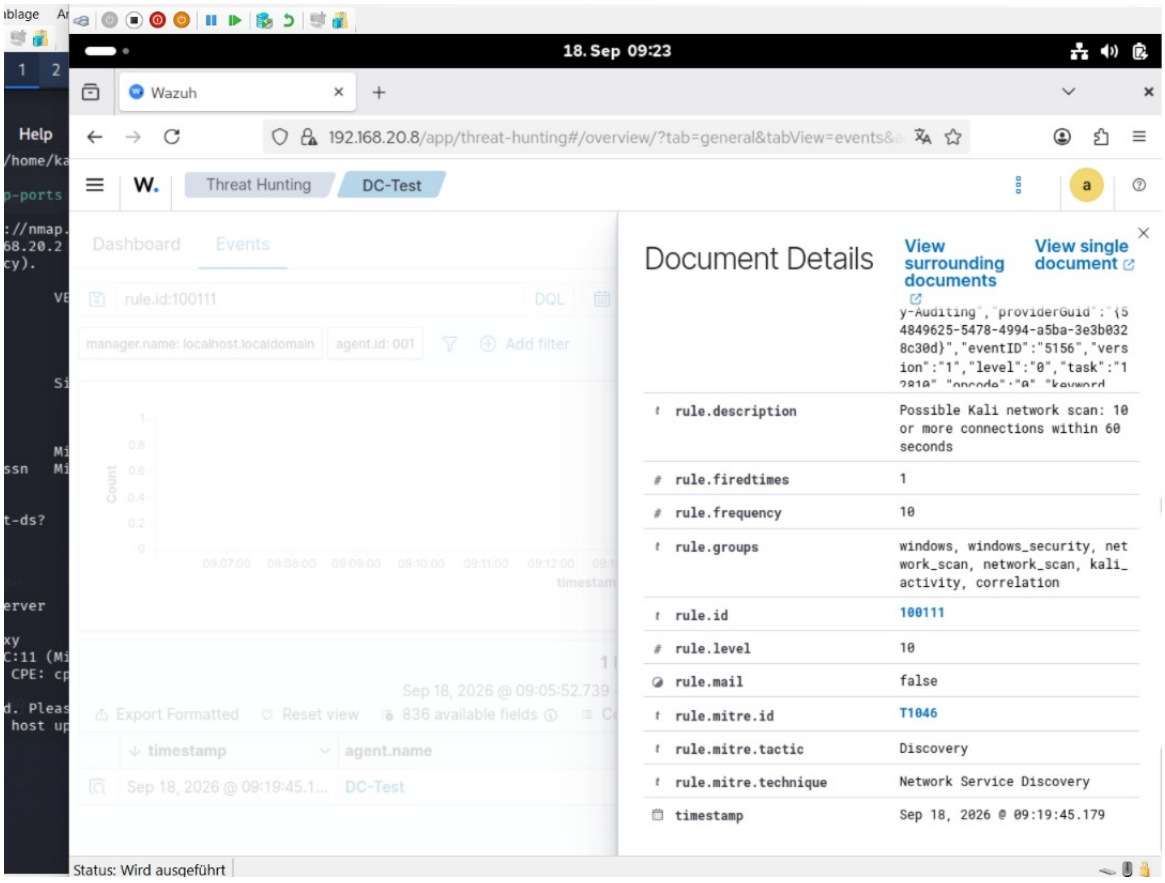


Figure 7: Correlated Nmap alert of rule 100111 to level 10

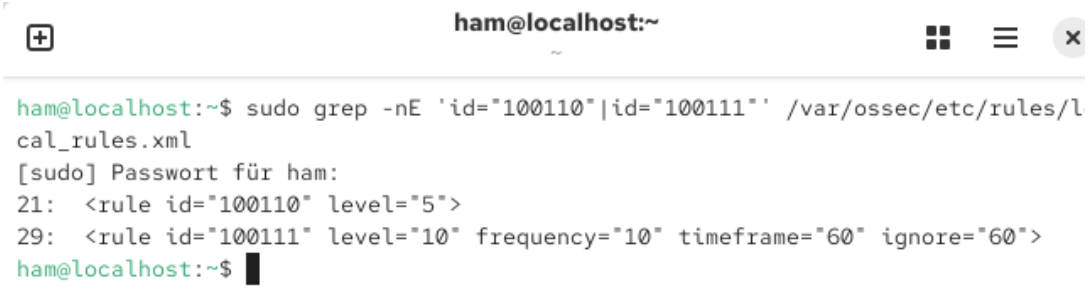


Figure 8: Successful validation of local rules

Production advice The solid Kali-IP is suitable for the reproducible laboratory test, but not as a general productive scan detection. A productive rule requires behavioral criteria, exceptions and false positive tests.

8. Ubuntu Hardening and Risk Acceptance

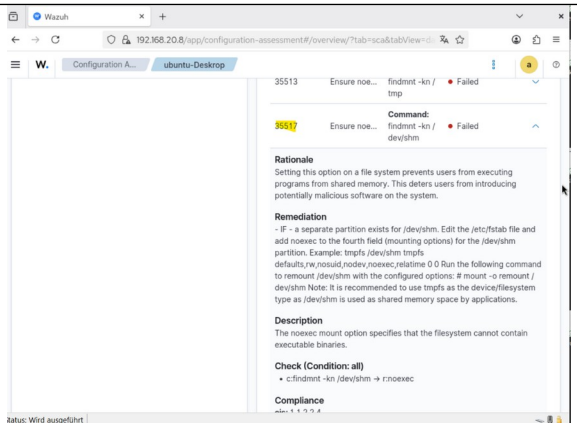
Improve safety value without distorting system function

SCA evaluation

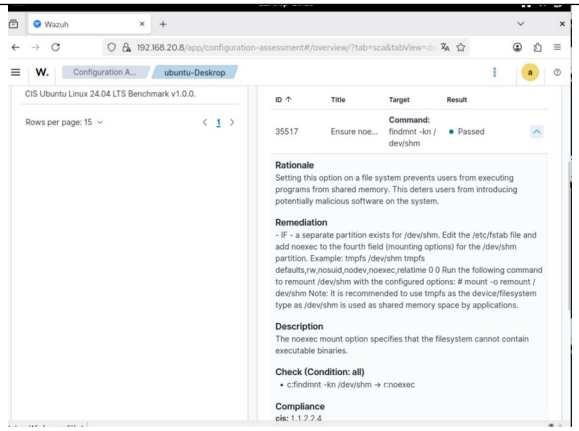
The CIS-based Wazuh-SCA-policy was evaluated at Ubuntu. The baseline value was 50 percent. Two tests were technically deepened: 35517 for the mount options of /dev/shm and 35509 for unneeded file system modules.

Check 35517: successfully hardened

The noexec option was added for /dev/shm. After re-mounting and restarting, the options rw,nosuid,nodev,noexec,relatime are confirmed. The check changed to Passed and the persistence has been checked.



Before: Check 35517 not met



After: Check 35517 passed after restart

Check 35509: justified exception

Unnecessary modules were blocked. FUSE remained active because the Ubuntu-Desktop requires it via GVFS. The fat module was fixed in the kernel and could not be unloaded like a loadable module. As the policy requires all sub-conditions, the overall check remained on Failed. The decision was documented as a risk acceptance instead of changing the policy only for a better percentage.

Security Judgment A green compliance value is not an end in itself. The technical dependence, operational impact and residual risk must be assessed together.

9. Sysmon and Encoded PowerShell

New detection chain at Windows-11 endpoint ham1

Objective and outcome

On 14.10.2026, Sysmon was installed at ham1 and DC-Test and was included in the Wazuh monitoring.The full detection test was performed at ham1. A non-hazardous base64-encoded PowerShell only issued the text SOC test. Sysmon recorded the process start, Wazuh processed the event and the integrated rule 92057 generated a level-12 alert.

Test point	Confirmed result
Endpoint	ham1 / Agent 002 / 192.168.20.5
Data source	Microsoft-Windows-Sysmon/Operational
Local event	Sysmon Event ID 1 with powershell.exe and EncodedCommand
MITRE allocation	T1059.001 – PowerShell
Wazuh detection	Rule 92057, level 12
Alert time	14.10.2026, 13:21:55 p.m.
Rule decision	Existing rule used; no double local rule created

Validated processing chain

1. The Wazuh Agent reads the Sysmon event channel.
2. Sysmon Event ID 1 contains process, command line and MITRE tag.
3. The manager processes the event completely and assigns it to Agent 002.
4. The integrated rule 92057 creates the alert in the dashboard.
5. The temporarily activated JSON total logging was deactivated after the control.

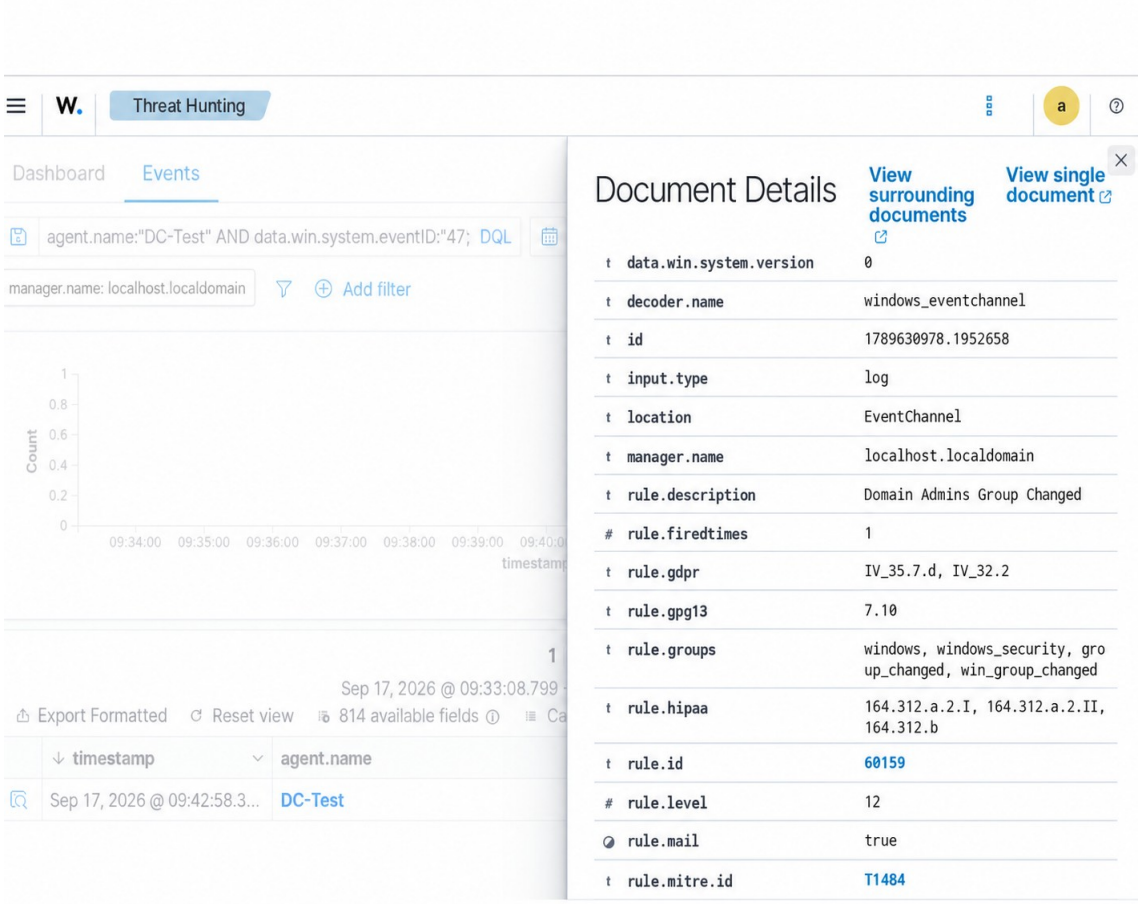


Figure 9: Wazuh Rule 92057 at level 12 with MITRE T1059.001

Technical evidence

Detection file	Contents
Windows11_Sysmon_EventID1_EncodedCommand	Local Sysmon process start with EncodedCommand
Wazuh_Sysmon_EventID1_Document_Details_T1059_001	Complete data set and MITRE assignment
Wazuh_PowerShell_encodedCommand_Rule_92057_Level12	Dashboard alert rule 92057 at level 12

Detection decision A separate rule would be technically possible, but technically unnecessary.The existing Wazuh rule already covers the use case. This avoids double alerts and unnecessary maintenance.

10. Alert Triage and Incident Response

From hit to reasoned decision

Triage run

1. Validate Alert: correct agent, right time, appropriate event or rule ID.
2. Collect context: user, source address, destination, process, command line and follow-up activity.
3. Check authorization: planned test, admin activity or unknown activity.
4. Classify: True Positive, False Positive, Benign True Positive or Incident.
5. Select reaction: document, further examine, contain or escalate.
6. Control completion: Cleanup, retest and lessons learned.

Short runbooks

Alert type	Important tests	Possible reaction
Network Scan	Source, destination ports, time window, approval, follow-up activity	escalate and limit access at unknown source
Failure to register	User, source IP, frequency, successful follow-up	Check account and source; lock or escalate samples
Privileged AD Group	4728/4729, executive account, change release	Remove unauthorized membership and secure account
PowerShell / Sysmon	Command line, parent process, user, host, network activity	Investigate process chain and persistence in unknown activity
FIM	Path, action, hash/Diff, user, approved change	Secure file, clear origin, isolate in case of manipulation

Example: Nmap Alert

Rule 100111 was rated as True Positive and Authorized Activity: Source 192.168.20.7 was the known potash test system, target was DC-Test and the time window was planned. Therefore, no isolation was necessary. The same alert from unknown source would have triggered an investigation and possible containment.

11. Results, Limitations, and Lessons Learned

Show technical strength without concealing laboratory boundaries

Results achieved

- Central monitoring of Windows Server, Windows 11 and Ubuntu in a common SIEM.
- Successful standard detections as well as self-developed and tested correlation for the network scan.
- MITRE traceable allocation for T1046 and T1059.001.
- Documented analyst decisions instead of purely technical alert collection.
- Clean final controls and removal of temporary test objects.

Current limits

Boundary	Meaning	Planned improvement
Small laboratory environment	No statement on productive load or 24/7-Betrieb	Measurable KPIs and longer test periods
A single Suricata sensor	The sensor sees traffic to Ubuntu-VM, but not a complete virtual switch traffic.	Later, dedicated sensor or controlled port Mirroring in the pure laboratory network test.
Limited Detection Collection	Only selected use cases tested	Additional Sigma rules and realistic attack and defense scenarios
No high availability	Single Points of Failure Remain	Backup, restore and operation documentation
Screenshots contain laboratory data	Check before public publication	Editing and secure GitHub selection

Lessons Learned

- An event is not yet a detection; a detection is not yet an incident.
- In the absence of alerts, auditing, transport, decoding and regulations must be examined separately.
- Fixed time base, unique agent names and stable addresses save a lot of analysis time.
- Own rules need validation, controlled tests and protection against double alerts.
- Risk Acceptance is better than a cosmetic change that endangers the operation.

12. Portfolio Structure for Job Applications

Documentation as entry point, technical artifacts separately and verifiable

Recommended GitHub structure

Path	Contents
README.md	Short description, architecture, main results and screenshots
docs/portfolio	This complete project documentation as PDF
docs/evidence	Selected, tested and possibly blackened proofs
rules/wazuh	Local rules 100110, 100111, 100120 and 100121 with explanation and test reference
configs	Adjusted configuration statements for agents, Sysmon and SCA
runbooks	Short runbooks for scan, login, AD, PowerShell and FIM
tests	Reproducible test cases, expected result and actual alert
rules/sigma	Validated Sigma rules with status, logsource, detection and test note
rules/suricata	Local Suricata rules 1000001 and 1000002 including tuning

Dealing with commands

The portfolio documentation only shows short, explanation-relevant technical information. Complete commands and configurations belong in separate, versioned files in the repository. Thus, the document remains readable, while recruiters and technical interviewers can still reproduce the implementation.

Publish check Before uploading passwords, tokens, internal data and unnecessary usernames are removed. Only commands and rules tested in the laboratory are published.

13. Development Roadmap

From a functioning laboratory to a convincing SOC portfolio

Phase	Objective	Status
1. Wazuh + Sysmon + Sigma	Sysmon at ham1 and DC-Test; Sigma rule created and validated	Completed
2. Suricata	Network telemetry integrated, scan detected, alarm flood reduced and Wazuh rule 100121 created	Completed
3. Attack/Verteidigung	Test realistic, safe scenarios reproducibly	Next focus
4. Dashboards and IR	Improve Triage, Prioritization, Key Figures and Playbooks	Runbook base available
5. GitHub Portfolio	Professionally publish rules, tests, proofs and documentation	Preparation
6. Application	Deriving CV project text and interview questions from the project	After that

14. Evidence Register

Evidence from the entire project stand until 17.10.2026

Core evidence in the document

No.	Evidence
01	Architecture and data flow
02	Three active Wazuh Agents
03	Ubuntu FIM with content change
Completed	Windows Event 4625 at ham1
Completed	Rule 60159 for Domain Admins
Next focus	Rule 100111 for Network Scan, Level 10
07	SCA Check 35517 before curing
08	SCA Check 35517 after restart Passed
09	Wazuh services active
10	Validation of rules Exit Code 0
11	AD test account removed
12	Sysmon Event ID 1 with EncodedCommand
13	Wazuh document details with MITRE T1059.001
14	Wazuh Rule 92057, level 12
15	Sysmon Event ID 1 for whoami /priv at DC-Test
16	Sigma rule proc_creation_win_whoami_priv without syntax and condition errors
17	Suricata TCP-SYN detection prior to tuning with 205 Wazuh-treffen
18	Suricata TCP-SYN detection after tuning with a Wazuh response
19	Wazuh Rule 100121, Level 8, MITRE T1046
20	Wazuh Dashboard: Rule 100121 with two hits

Documentation principle

The evidence shows only successful, relevant results. Mistry was used for problem solving, but not taken as an unnecessary step-by-step history. Where an error led to an important technical realization – for example, in the Nmap detection gap or in the SCA exception – the cause is documented with the final solution.

15. Detection Engineering with Sysmon, Sigma, and Suricata

Sysmon and Sigma and integrate Suricata as a new network data source

15.1 Objective and overall result

This project section combines host-based process monitoring, portable detection logic and network-based telemetry to a consistent analysis process. Sysmon recorded a controlled whoami /priv process start to DC-Test. The associated Sigma rule was checked with Sigma CLI. Suricata was installed at ubuntu-Desktop, integrated into Wazuh and validated with an authorized SYN scan of Kali. Threshold tuning reduced the number of hits from 205 to a meaningful alert. A separate Wazuh rule raised the final hit to level 8 and assigned it to MITRE T1046.

Block	Result
Sysmon	Process Create successfully recorded at DC-Test for whoami /priv
Sigma	Rule proc_creation_win_whoami_priv.yml technically validated
Suricata	ICMP and TCP-SYN activity detected in eve.json
Wazuh integration	Suricata-JSON processed as a local log source
Alert tuning	205 similar hits reduced to one alert per source and time window
MITRE ATT&CK	Final rule 100121 at level 8 with T1046 Network Service Discovery

15.2 Sysmon and Sigma

Sysmon was installed on Windows 11 and Windows Server. For the new test, the non-hazardous command whoami /priv was executed on DC-Test. Sysmon Event ID 1 provided process name, full command line, user and parent process context. The existing Wazuh rule 100120 confirmed end-to-end processing. The detection was then saved as a Sigma rule to make logic understandable, versionable and later transferable to other SIEM platforms.

Important test and test commands:

```
whoami /priv

Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 10 |
  Where-Object {$_.Message -match "whoami.exe"} |
  Select-Object -First 1 TimeCreated, Id, Message

sigma check ~/soc-portfolio/rules/sigma/proc_creation_win_whoami_priv.yml
```

Validation result The Sigma test did not report rule errors or condition errors. The rule status was performed as experimental for a newly developed laboratory rule. Thus, the rule is technically clean without presenting it as ready for production.

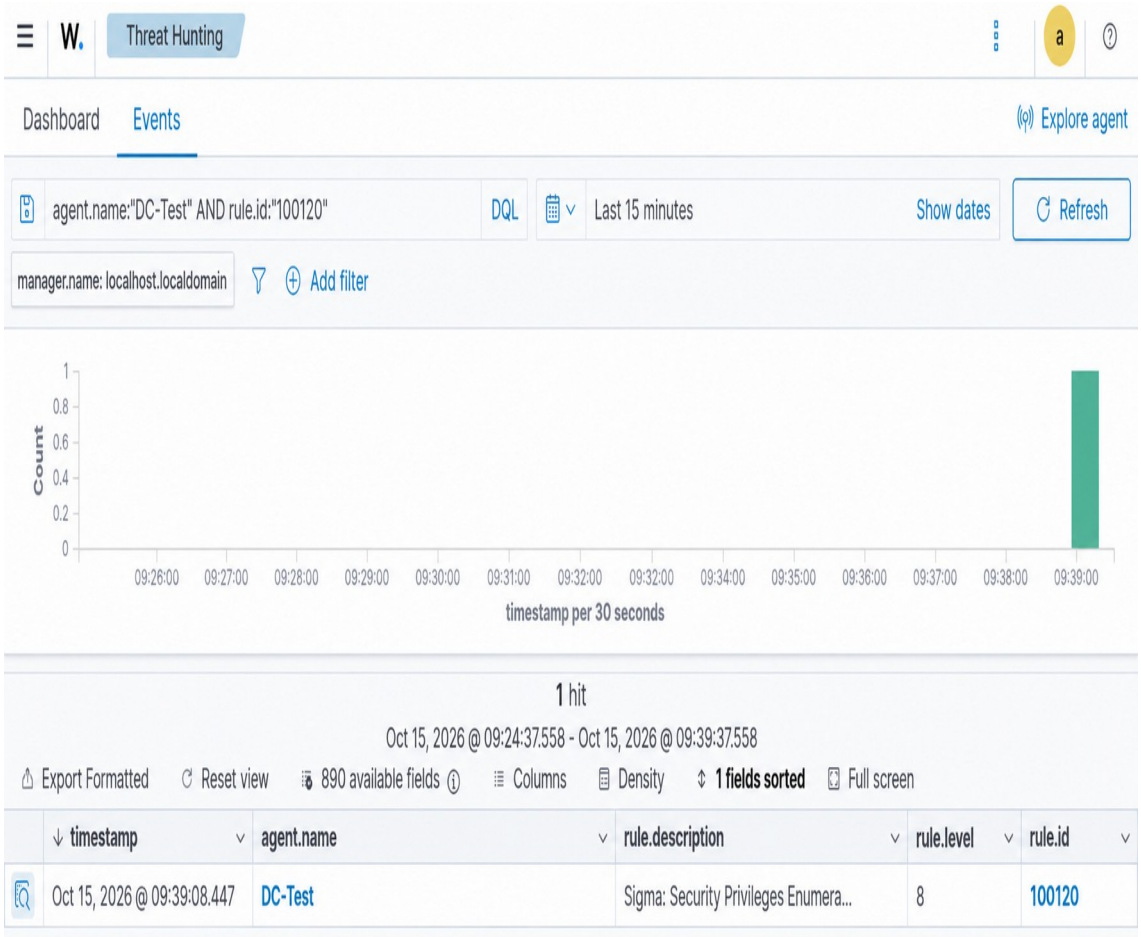


Figure 10: Wazuh Rule 100120 for whoami /priv to DC-Test

15.3 Suricata install and configure

Suricata has been installed directly on ubuntu-Desktop. eth0 is monitored by the sensor at 192.168.20.20. HOME_NET has been limited to the internal laboratory network 192.168.20.0/24. The physical Hyper-V host has not been altered and has not been used as an attack target.

Installation and basic testing:

```
sudo apt-get install software-properties-common -y
sudo add-apt-repository ppa:oisf/suricata-stable -y
sudo apt-get update
sudo apt-get install suricata jq -y

sudo suricata-update
sudo suricata -T -c /etc/suricata/suricata.yaml
sudo systemctl restart suricata
sudo systemctl is-active suricata
```

Configuration	Value
HOME_NET	[192.168.20.0/24]
Capture interface	Eth0
Event Output	/var/log/suricata/eve.json
Standard rules	/var/lib/suricata/rules/suricata.rules
Local rules	/var/lib/suricata/rules/local.rules

15.4 Suricata connect to Wazuh

The Wazuh Agent reads Ubuntu as a JSON log source. Thus, the Suricata fields such as source address, destination address, signature ID, signature text and Severity remain searchable in a structured manner.

```
<localfile>
 <log_format>json</log_format>
 <location>/var/log/suricata/eve.json</location>
</localfile>
```

```
sudo systemctl restart wazuh-agent
sudo systemctl is-active wazuh-agent
```

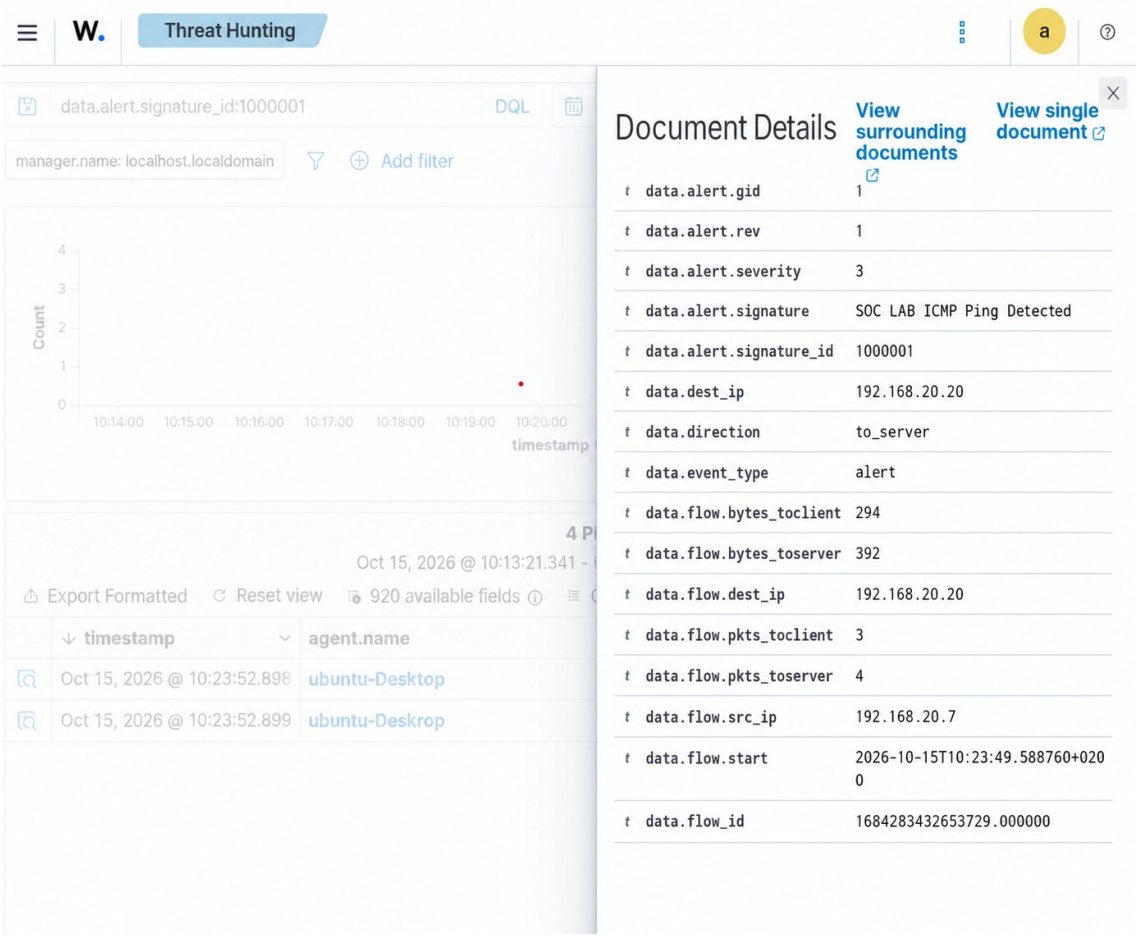


Figure 11: Suricata ICMP SID 1000001 structured in Wazuh

15.5 Local Suricataity rules and controlled tests

First, an ICMP rule with SID 1000001 confirmed that packet capture, Suricata analysis and Wazuh forwarding work. For the scan-use case, SID 1000002 was then created. The rule detects TCP-SYN packets against Ubuntu and limits the output to one alert per source address within ten seconds.

```
alert icmp any any -> 192.168.20.20 any (msg:"SOC LAB ICMP Ping Detected"; itype:8; sid:1000001; rev:1;)
```

```
alert tcp any any -> 192.168.20.20 any (msg:"SOC LAB Possible TCP SYN Port Scan"; flags:S; flow:stateless; threshold:type both, track by_src, count 20, seconds 10; classtype:attempted-recon; sid:1000002; rev:1;)
```

Authorized test for Kali Linux:

```
sudo nmap -sS -Pn -T3 -p 1-1000 192.168.20.20
```

Local check for Ubuntu:

```
sudo jq -c 'select(.event_type=="alert" and .alert.signature_id==1000002) |
{timestamp,src_ip,dest_ip,dest_port,signature_id:.alert.signature_id,signature:.alert.signature,severity:.alert.
severity}' /var/log/suricata/eve.json | tail -n 5
```

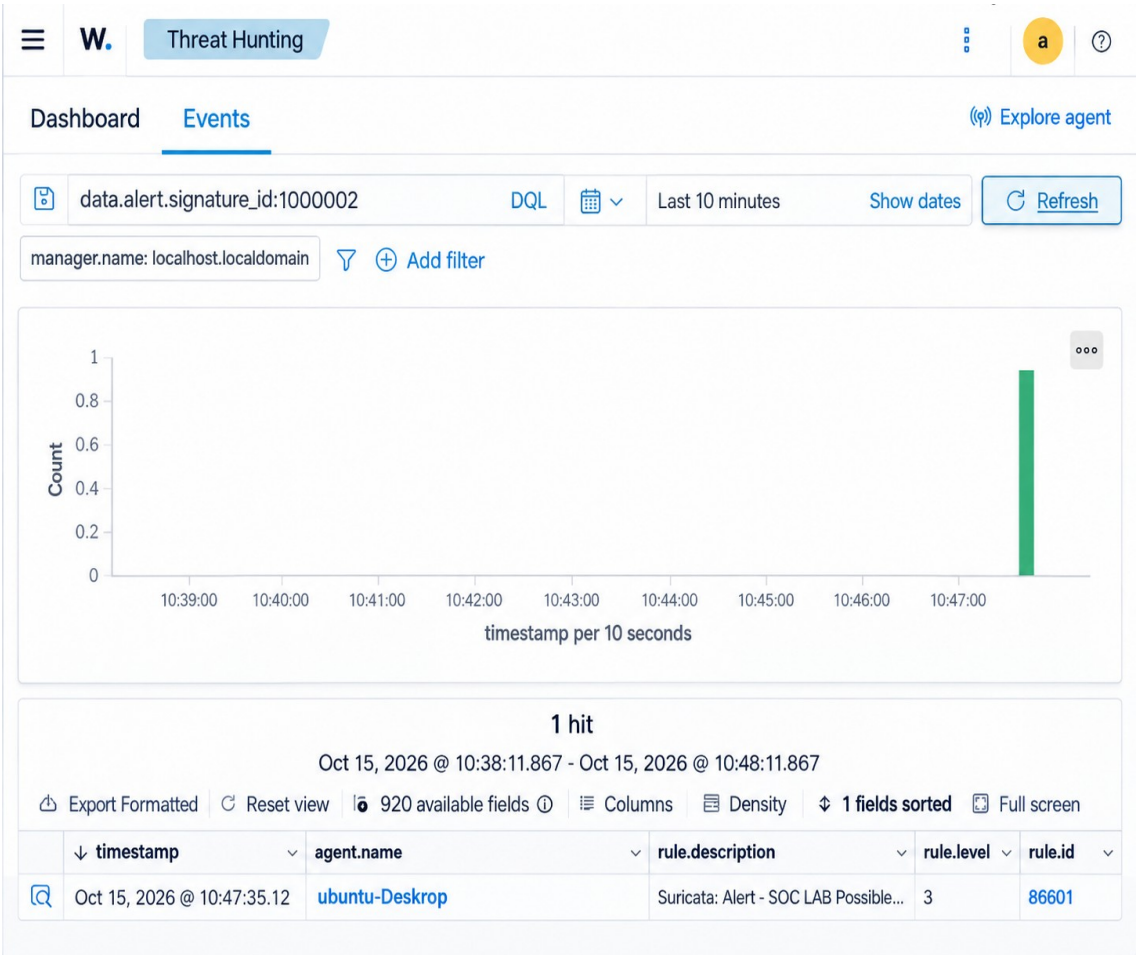


Figure 12: After tuning, an Wazuh result for Suricata SID 1000002 remains.

15.6 Wazuh rule and MITRE assignment

The generic Suricata Rule 86601 initially generated a Level 3 alert. For the project-specific scan use case, Wazuh Rule 100121 was added on the manager. It matches only Suricata SID 1000002, raises the alert to Level 8, and maps the behavior to MITRE ATT&CK T1046 Network Service Discovery.

```
<group name="suricata,network_scan,">
 <rule id="100121" level="8">
  <if_sid>86601</if_sid>
  <field name="alert.signature_id">^1000002$</field>
  <description>Suricata: TCP SYN port scan detected against Ubuntu sensor</description>
  <mitre><id>T1046</id></mitre>
 </rule>
</group>
```

```
sudo /var/ossec/bin/wazuh-analysisd -t
sudo systemctl restart wazuh-manager
sudo systemctl is-active wazuh-manager
```

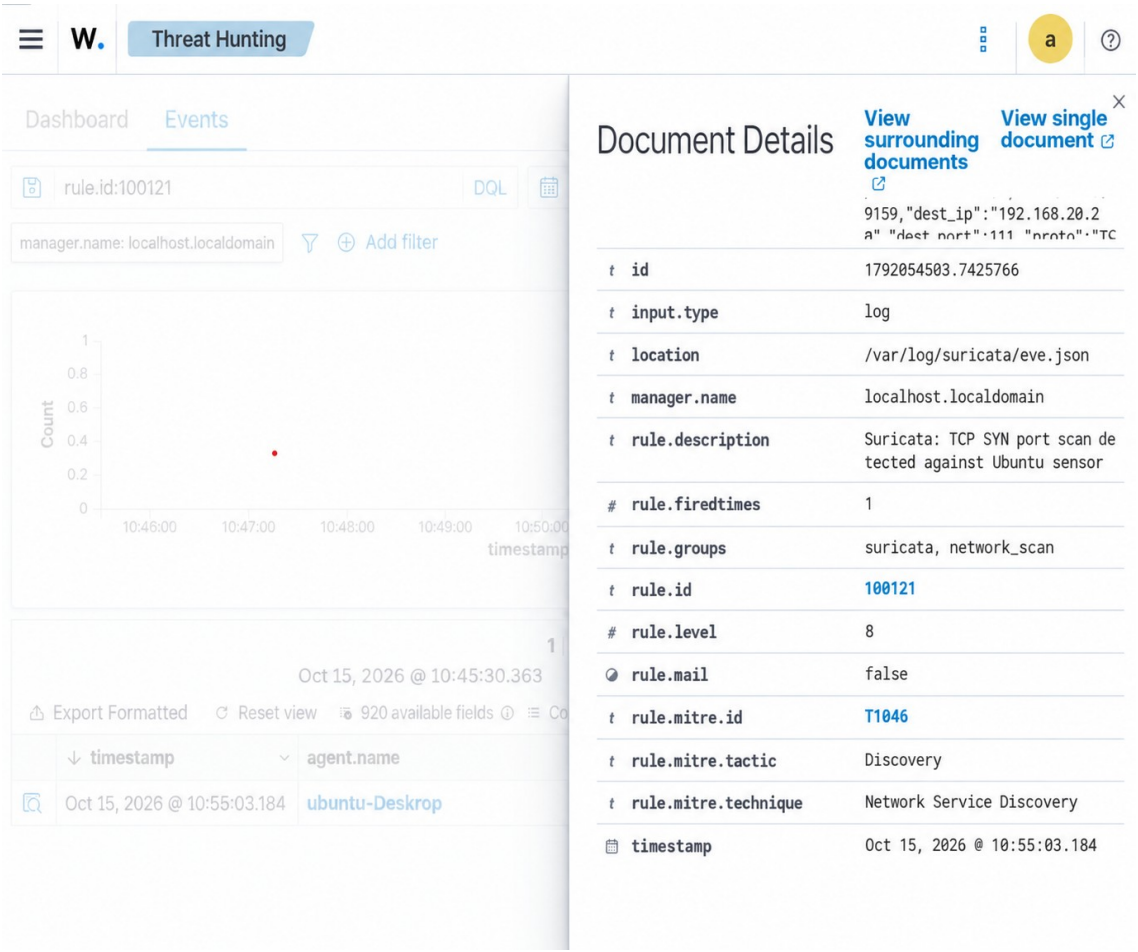


Figure 13: Finale Wazuh Rule 100121 at level 8 with MITRE T1046

15.7 Analyst evaluation and next phase

The alert is a True Positive because the TCP SYN scan actually occurred. It was also an authorized laboratory activity and therefore not an incident. During Triage, the source 192.168.20.7, target 192.168.20.20, signature 1000002, time window, Rule ID 100121, Level 8, and MITRE T1046 mapping were reviewed.

Complete Phases 3 and 4 The documented case studies combine controlled tests, telemetry, alert triage, classification and incident response decisions. The next focus is structured publication as GitHub portfolio.

15.8 Alert analysis on the Wazuh rule 100121

On 16.10.2026, the threat hunting module in the dashboard was searched for rule.id:100121. For the visible period "Last 30 days" the view showed two alert events. The event list showed a hit from 15.10.2026 at 10:55 and one from 16.10.2026 at 08:32. These two documents prove two captured alerts; from this it is not derived that two separate scan runs took place.

The detailed views opened confirmed Agent ubuntu-Desktop (ID 003, 192.168.20.20), source Kali 192.168.20.7, target 192.168.20.20, protocol TCP, Suricata signature SOC LAB Possible TCP SYN Port Scan (SID 1000002) and Wazuh rule 100121 at level 8. The older Alert of 15.10.2026 recorded target port 111, the newer of 16.10.2026 target port 199. This means that two alert documents are available for different flows; this cannot be safely concluded for two separate scanning runs. The data.alert.action field showed that both details were allowed: Suricata recognized traffic, but did not block it.

Time stamp	Source → Objective	TCP target port	Action
15.10.2026 10:55:03.184	192.168.20.7 (Kali) → 192.168.20.20 (Ubuntu)	111	allowed – detected, not blocked
16.10.2026 08:32:09.683	192.168.20.7 (Kali) → 192.168.20.20 (Ubuntu)	199	allowed – detected, not blocked

Both alerts belong to Suricata SID 1000002 and Wazuh-rule 100121 (level 8). The two alert documents prove different flows, but not necessarily two separate scans.

Rating: Detection has been successful. The display of zero alerts from level 12 and above is compatible with Rule Level 8 and does not mean a failed detection. Due to the approved laboratory test, the classification is True Positive – Authorized Security Test; no containment was required.

16. SOC Case Study: Suricata TCP SYN Port Scan

From controlled SYN scan to documented analyst decision

16.1 Objective and procedure

This case study converts the already validated network detection into a complete SOC workflow. An authorized Kali TCP-SYN scan was detected by Suricata, structured to Wazuh, linked to MITRE ATT&CK and then triagated. The test remained completely within the isolated laboratory network 192.168.20.0/24.

Detection Chain Kali Nmap -> Suricata SID 1000002 -> eve.json -> Wazuh Rule 100121 -> MITRE T1046 -> Triage -> Closure.

16.2 Case data

Field	Evaluation
Case ID	SOC-2026-001
Title	Network Discovery detected from Kali
Source	192.168.20.7 - Kali Linux
Objective	192.168.20.20 - Ubuntu / Suricata Sensor
Test	sudo nmap -sS -Pn -p 1-1000 --reason

Field	Evaluation
	192.168.20.20
Suricata	SID 1000002 - SOC LAB Possible TCP SYN Port Scan
Wazuh	Rule 100121, level 8
MITRE ATT&CK	T1046 - Network Service Discovery
Classification	True Positives - Authorized Security Test
Status	Closed
Dashboard query	rule.id:100121; last 30 days: 2 alert events (15. and 16.10.2026); target ports 111 and 199
Suricata action	allowed; traffic detected was not blocked

16.3 Triage and Analyst Decision

1. Validate alert: rule ID, Severity, timestamp, source and destination address as well as the Suricata signature have been checked.
2. Source: 192.168.20.7 is the known potash system of the isolated test laboratory.
3. To determine scope: The scan was directed exclusively against 192.168.20.20 and the ports 1 to 1000.
4. Check context: Time and scanning parameters match the authorized test.
5. Check accompanying activity: There was no additional evidence of exploitation, persistence or data outflow.
6. Deciding: Detection is technically correct and therefore a true positive; the confirmed laboratory test is not a security incident.

Analyst rating The TCP-SYN scan actually took place and was correctly detected. Since the source, target, time window and purpose of the test were known and approved, the alert was closed as True Positive - Authorized Security Test. A false positive would be technically wrong because the reported behavior was real.

16.4 Incident Response Runbook

Phase	Analyst Action	Escalation criterion or result
1. Validates	Confirm alert fields, timestamps, rules, signature and data source.	Returns invalid or incomplete data for technical testing.
2. Scope	Determine source, destination, port range, frequency and other alerts of the same source.	Increase priority for multiple targets or repeated activity.

Phase	Analyst Action	Escalation criterion or result
Enrich	Check asset role, owner, change or test release and known scanners.	Treat unknown source or lack of release as suspicious.
4. Classify	Set True Positives, False Positives, or Authorized Benign True Positives.	Escalate as potential incident with unknown motivation.
5. Contain	In case of unauthorized scan, isolate or block source and monitor affected systems.	In the confirmed laboratory test, no containment was required.
6. Close	Document the decision, evidence, measures and improvements in a comprehensible manner.	Set Status Closed with justification and reference to Evidence.

16.5 Completion and improvements

For a productive operation, the same detection would escalate at an unknown source, multiple target systems or additional exploit activity. The Threshold tuning already implemented reduces alert duplicates without overlooking the actual scan. Figure 12 for the tuning result and Figure 13 for the final Wazuh rule with MITRE assignment serve as evidence. Section 15.8 contains the documented evaluation of the two alert events in the 30-day period.

Closure Note Authorized laboratory scan confirmed. Detection worked as designed. No compromise indicators identified. No containment required. Case closed.

17. SOC Case Study: Encoded PowerShell

From Sysmon telemetry to context-based incident decision

17.1 Objective and procedure

This case study uses the already successfully executed EncodedCommand test on Windows 11. Sysmon captured the process chain and the full command line. Wazuh recognised the Base64-encoded PowerShell behaviour with the integrated rule 92057 and assigned it to MITRE ATT&CK T1059.001. For the case study, the test was not re-executed; the existing validated evidence was analysed.

Detection Chain Windows 11 ham1 -> PowerShell EncodedCommand -> Sysmon Event ID 1 -> Wazuh Rule 92057 -> MITRE T1059.001 -> Triage -> Closure.

17.2 Case data

Field	Evaluation
Case ID	SOC-2026-002
Title	Encoded PowerShell detected on Windows 11
Endpoint	ham1 - Windows 11, 192.168.20.5
User context	Authorized local laboratory test
Data source	Microsoft-Windows-Sysmon/Operational, Event ID 1
Detection feature	powershell.exe with -EncodedCommand and Base64 content
Wazuh	Rule 92057, level 12
MITRE ATT&CK	T1059.001 - PowerShell
Classification	True Positives - Authorized Security Test
Status	Closed

17.3 Triage and Analyst Decision

1. Validate alert: check host, user, time stamp, rule ID, level and MITRE assignment.
2. Check process chain: Evaluate image, parentimage, commandline and running user from Sysmon Event ID 1.
3. Decode EncodedCommand: Decode the Base64 content offline as Unicode and evaluate the actual PowerShell command without re-running it.
4. Determine Scope: Search for the same command line, the same user, and comparable alerts for further endpoints.
5. Check accompanying activity: examine network connections, child processes, file changes, persistence, and credential accesses in the time window.

6. Deciding: The test command was authorized and safe; the detection is correct, but there is no security incident.

Analyst rating Encoded PowerShell is a strong suspicious feature, but no evidence of harmful activity on its own. In this case, host, time, user context and decoded content confirmed the approved laboratory test. The alert was therefore closed as True Positive - Authorized Security Test.

17.4 Incident Response Runbook

Phase	Analyst Action	Escalation criterion or result
1. Validates	Wazuh and Sysmon fields as well as completeness of the CommandLine confirm.	Provide missing telemetry for sensor or agent testing.
2. Decode safely	Decode Base64 content offline; do not execute the content unchecked.	Download, obfuscation, credential access or persistence immediately escalate.
3. Scope	Find the same commands, users, hosts, hashes and follow-up processes in the time window.	Multiple endpoints or repeated executions increase priority and scope.
4. Enrich	Check Asset Criticism, User Role and Change or Test Sharing.	Treat unknown user or missing release as incident.
5. Contain	Isolate host with confirmed malicious activity, stop process and secure artifacts.	In the authorized test, no containment was required.
6. Close	Document decoded content, decision, evidence and measures.	Closed as an authorized True Positive with reference to Figure 9.

17.5 Completion and improvements

In a production environment, the same detection would immediately escalate in unknown users, suspicious decoded content, external network connections, or follow-up processes. The decision is based not only on the alert name, but on process context, decoded content, and correlated activities. Figure 9 documents Rule 92057 on level 12 with MITRE T1059.001.

Closure Note Authorized Encoded PowerShell laboratory test confirmed. Sysmon telemetry and Wazuh detection worked as designed. Decoded content was benign. No compromise indicators identified. Case closed.

18. SOC Case Study: Accounts and Privileged Groups

Windows notification events and changes to Domain Admins jointly evaluate

18.1 Objective and technical demarcation

The case study summarizes already validated Windows security events into a common triage use case. Event 4625 documents a failed registration, Event 4624 a successful registration, Event 4728 adding a member to a security-enabled global group and Event 4729 its removal. The original laboratory tests were authorized individual tests. Therefore, no blank chain of attacks between the events is claimed; it is shown how an SOC would correlate such signals temporally and professionally.

Detection Chain DC-Test Security Log -> Wazuh Windows decoders and rules -> Authentication and group change analysis -> MITRE T1098.007 -> Triage -> Closure.

18.2 Case data

Field	Evaluation
Case ID	SOC-2026-003
Title	Account logs and Domain Admins membership change
System	DC-Test - Windows Server, 192.168.20.2, domain test.local
Event 4625	Failure to register
Event 4624	Successful application
Event 4728	Member added to a security-enabled global group
Event 4729	Member removed from a security-enabled global group
Wazuh	Rule 60159 for the monitored change to Domain Admins
MITRE ATT&CK	T1098.007 - Additional Local or Domain Groups
Classification	True Positives - Authorized Administrative Test
Status	Closed

18.3 Triage and Analyst Decision

1. Validate events: computer, time stamp, event ID, trading account, destination account and source address.

2. Evaluate logins: Check for 4625 error reason, status code, logon type and frequency; confirm with 4624 logon type and source device.
3. Perform correlation cautiously: Only with appropriate users, sources and time windows assume a connection between 4625 and 4624.
4. Investigate group change: Check for 4728 and 4729 trading account, affected member and target group Domain Admins.
5. Check authorization: Change clearance, administrator responsibility and purpose of group membership confirm.
6. Determine scope: Search for further account changes, privileged logons and identical actions on the domain controller.
7. Deciding: The logged actions were actually carried out, but were approved laboratory tests; there is no Security Incident.

Analyst rating The events are true positives because logins and group changes have been recorded in real terms. The classification as an authorized administrative test was only performed after checking users, timing, target group and test context. Repeated 4625 events can indicate Brute Force, but are not classified as MITRE T1110 only due to a single error attempt.

18.4 Incident Response Runbook

Phase	Analyst Action	Escalation criterion or result
1. Validates	Confirm event ID, host, user, source, time and Wazuh rule.	Technically clarify incomplete fields or time deviations.
Correlates	Connect 4625, 4624, 4728 and 4729 only using matching fields and time windows.	Do not construct a chain of attacks from events that are independent of time.
3. Verify change	Tune domain admins change with change sharing and administrator.	escalate missing release immediately as high priority.
4. Scope	Find other privileged logons, account and group changes as well as affected systems.	Multiple accounts or systems expand Incident Scope.
5. Contain	Remove member if unauthorized, lock account and check sessions.	In the confirmed laboratory test, no containment was required.
6. Recover	Validate permissions, reset passwords or tokens if required and increase monitoring.	Normal operation only after confirmed permission testing.
Close	Document evidence, authorization, decision and measures in a comprehensible manner.	Closed as an authorized True Positive.

18.5 Completion and improvements

For a productive operation, changes to highly privileged groups must be immediately compared with change data, responsible administrator and other login events. Especially important is the separation between actually correlated activities and only jointly viewed events. This keeps the analysis reproducible and avoids unresolved conclusions.

Closure Note Authorized authentication and Domain Admins membership tests confirmed. Windows Security events and Wazuh detection worked as designed. No unauthorized privilege change or compromise indicators identified. Case closed.

19. Dashboard Overviews and Presentation

Wazuh overview, focused alarm views and SOC dashboard

The following screenshots document the dashboard views for the daily overview and the targeted triage. The overall view shows key figures and time distribution. The filter on Rule 100121 narrows the search to two captured alerts; the detailed view shows the checked Suricata fields. The stored SOC overview board bundles agent, time, and Severity panels as well as a unique number for alerts of levels 12 to 15.

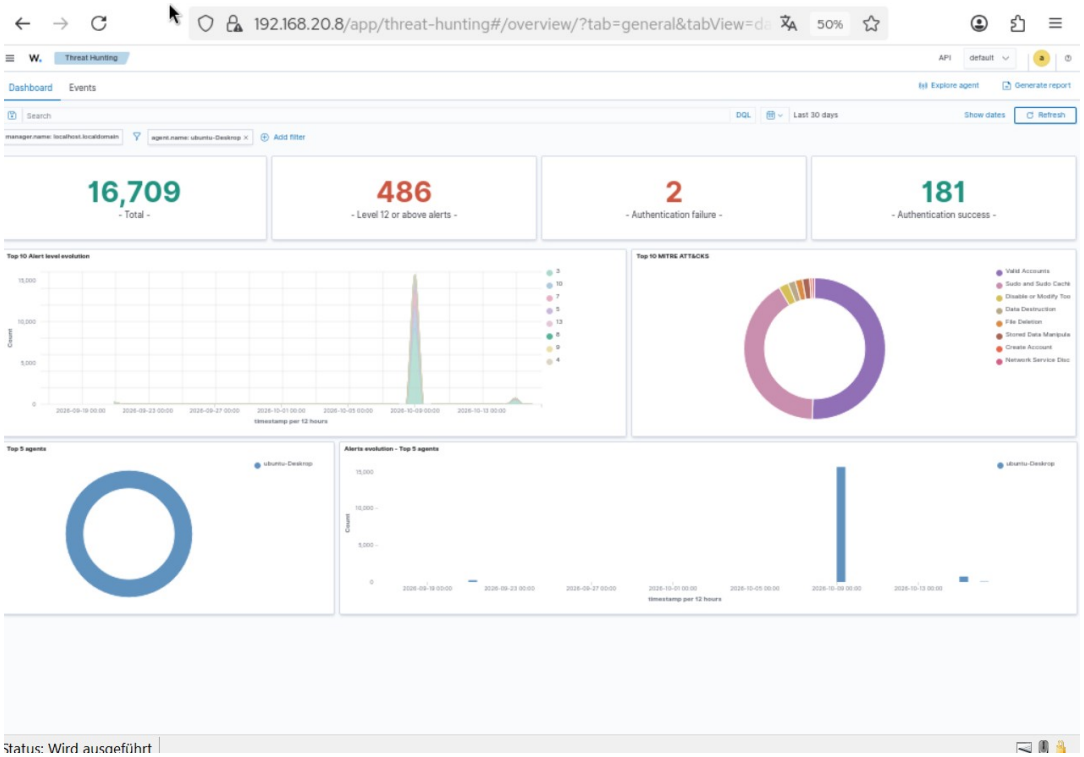


Figure 14: Wazuh Threat-Hunting overview with total number, level-12+ alerts and authentication key numbers

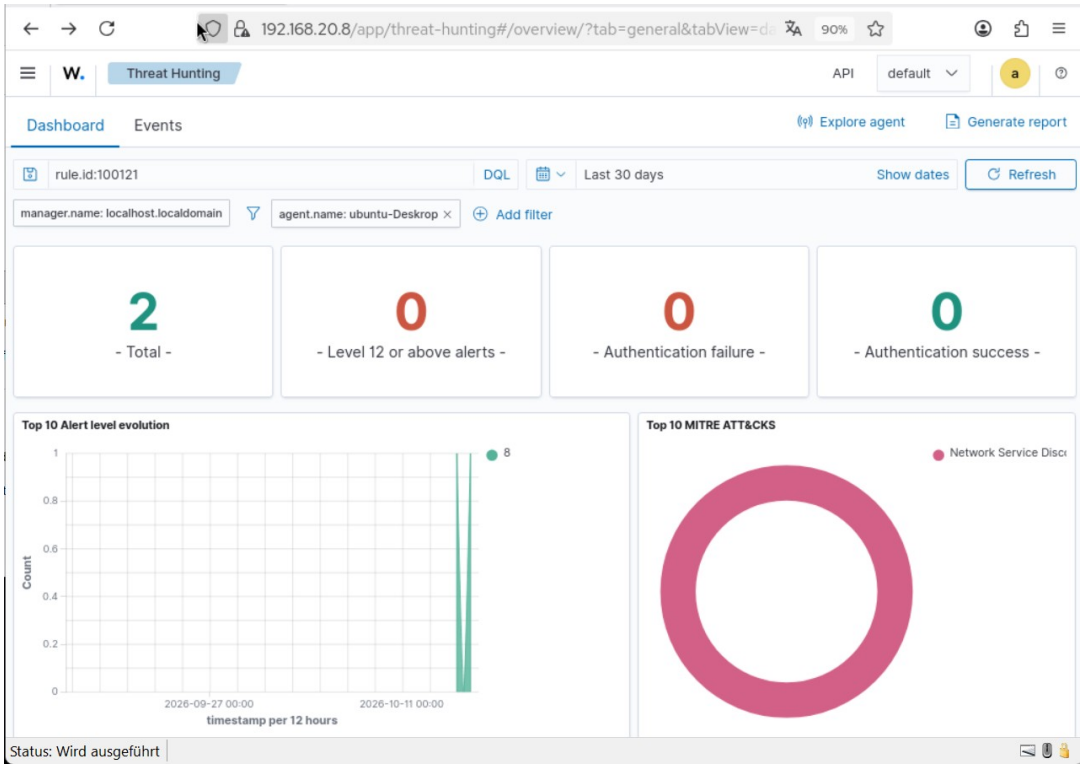


Figure 15: Dashboard with Rule-ID 100121 and two found alerts

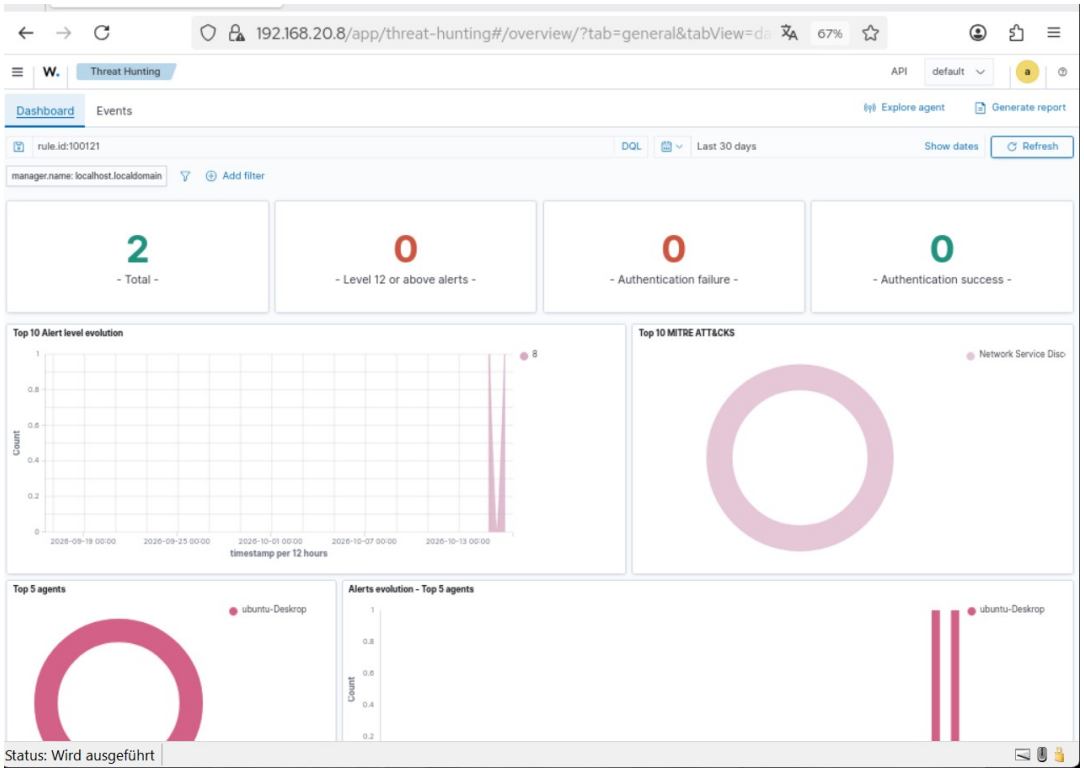


Figure 16: Filtered Rule-100121 view with key figures and alert distribution

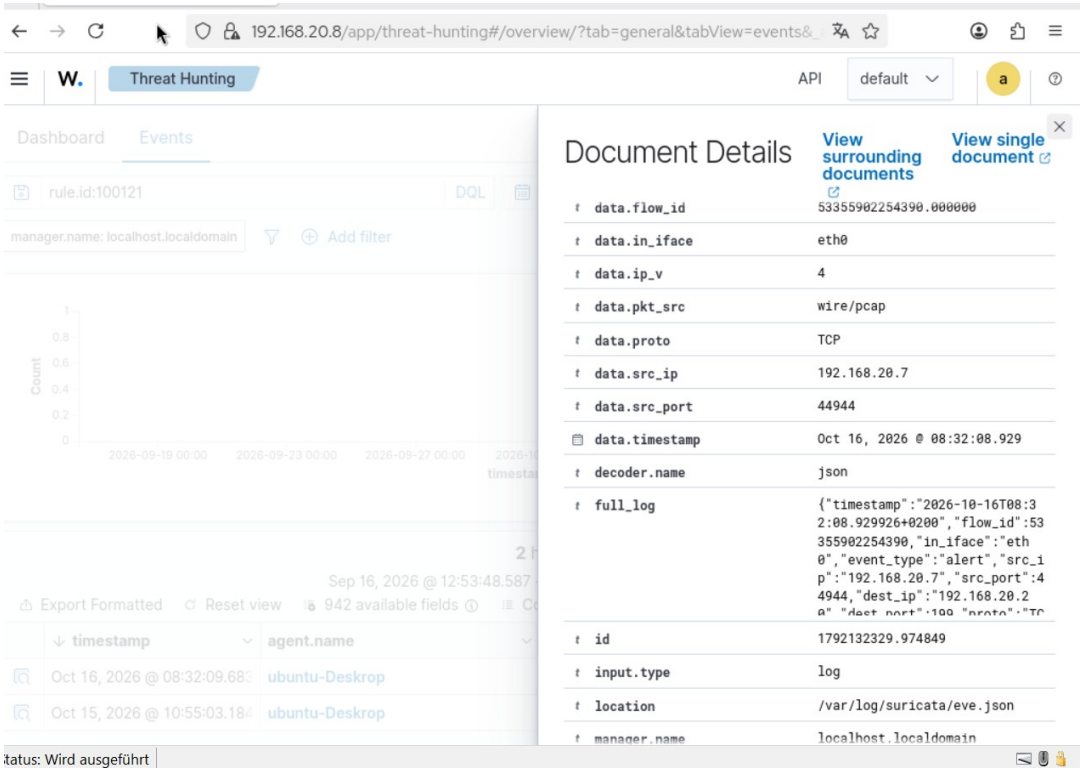


Figure 17: Detail fields of a Suricata-TCP-SYN scan alert in Wazuh

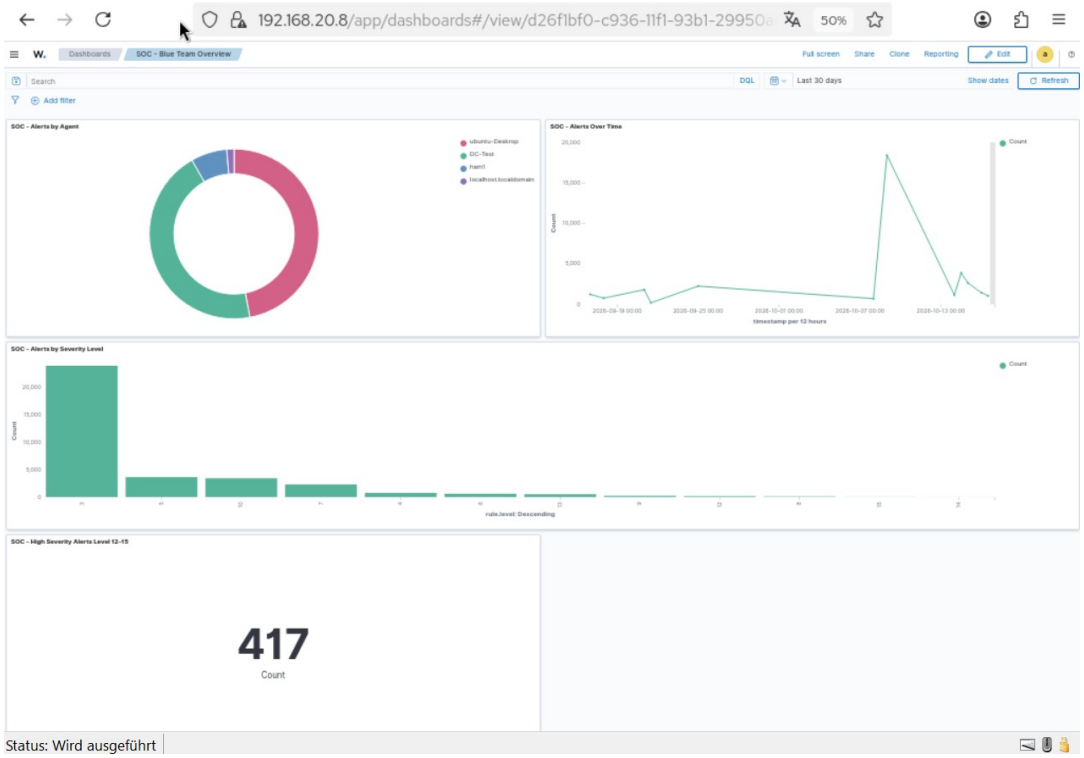


Figure 18: Stored SOC overview board with agent, time and Severity evaluation and 417 alerts of levels 12 to 15

Dashboard optimization result The board is saved and uses the period "Last 30 days" . The additional value "SOC - High Severity Alerts Level 12-15" shows 417 prioritized alerts and supports the quick selection of relevant events for the subsequent triage.

19.1 Prioritized Triage Overview

The SOC overview board has been extended by two evaluations for the daily alert triage. The table "SOC - Top 10 Alert Rules" shows the most frequently triggered rules. The table below "SOC - High Severity Triage Summary" limits the evaluation to Wazuh levels 12 to 15 and groups the hits by severity, rule ID, rule description and number.

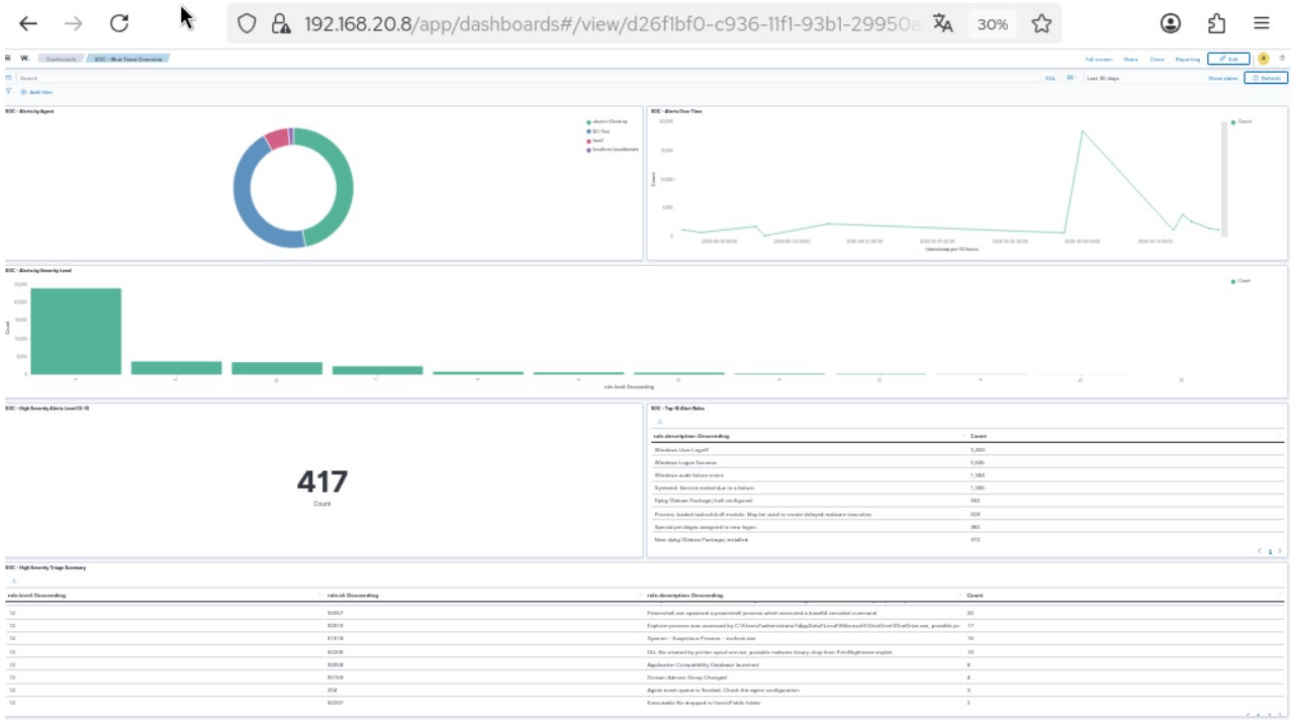


Figure 19: Stored SOC dashboard with top 10 rules and prioritized triage summary for alerts of levels 12 to 15

Triage Optimization Result The combined view allows quick prioritization: Rule level marks the severity, Rule ID makes the investigation reproducible, the description provides the technical context and the Count highlights frequently occurring alerts. Thus, an analyst can select eye-catching rules and then examine them in Threat Hunting until the single event.

19.2 Triage of an PowerShell alert

An existing high-severity alert at Windows-11 endpoint ham1 was examined in the Threat Hunting module up to event level. The aim was not to re-execute a test, but to evaluate a previously detected PowerShell event after SOC-Triage run.

Test point	Findings
Endpoint / user	ham1 / TEST\Administrator
Data source	Sysmon Event ID 1 - Process Creation
Wazuh detection	Rule ID 92057, level 12
Acknowledged process	PowerShell with Base64-encoded command
Decoded content	Write output "SOC test"
MITRE ATT&CK	T1059.001 - PowerShell; Tactics: Execution
Evaluation	True Positives - Authorized Test
Reaction	No containment required; analytically closed with justification

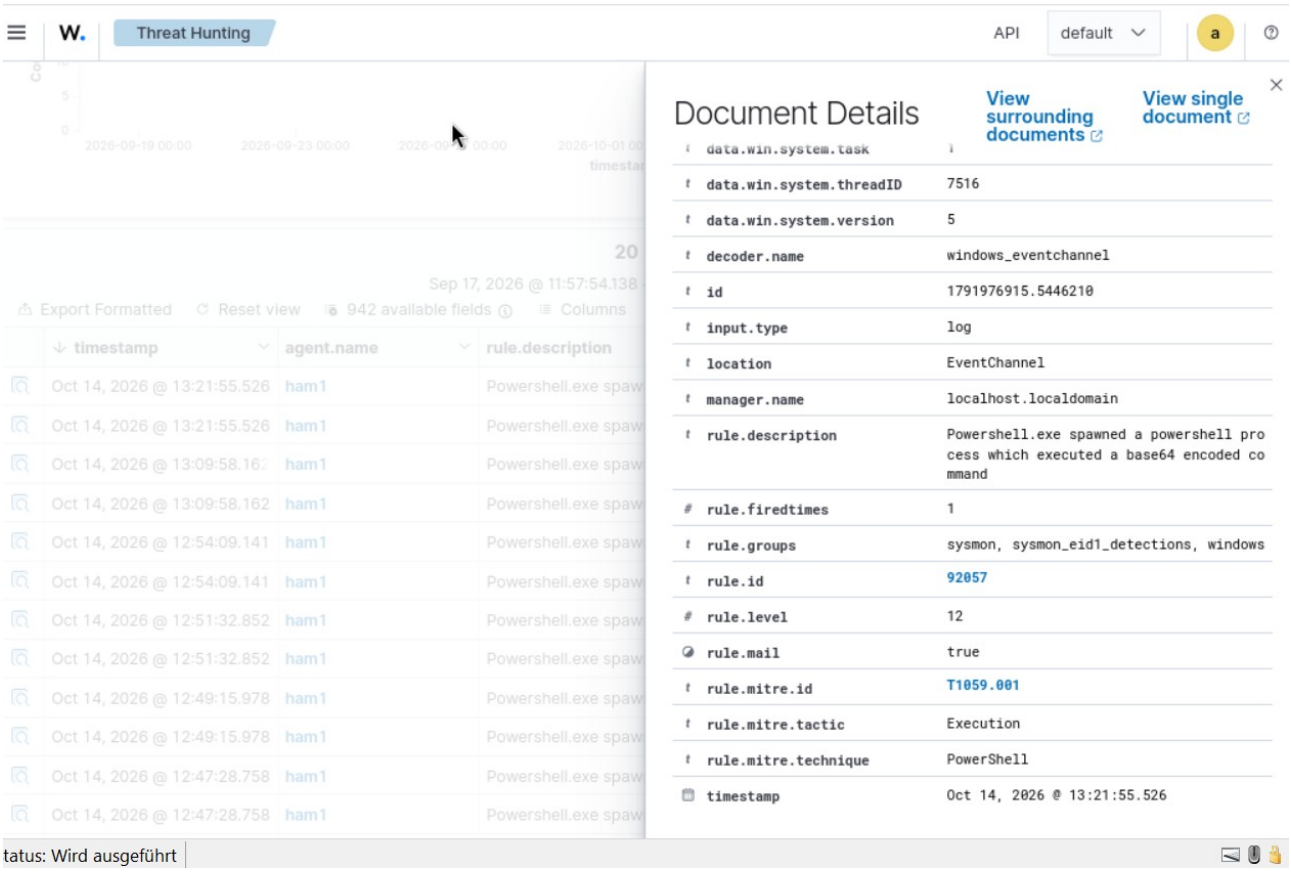


Figure 20: Wazuh-Triage evidence for rule 92057 with level 12, MITRE ATT&CK T1059.001 and the tactics Execution

Triage Decision The event is a true positive, because the coded PowerShell process was actually executed and correctly detected by Sysmon and Wazuh. However, the decoded content was a harmless, previously authorized laboratory test. There is no evidence of compromise or unauthorized activity. Therefore, neither isolation nor any other containment measures were required.

Closure Note Authorized PowerShell Base64 test confirmed on endpoint ham1. Sysmon Event ID 1 and Wazuh Rule 92057 (Level 12) detected the activity as designed. Decoded content was write output "SOC test". No malicious follow-up activity identified; no containment required.

19.3 Triage of an Suricata networkalert

The already documented Suricata alert was examined as the second triage case at network level. The basis were the two hits to Rule 100121 as well as the dashboard and event details shown in Figures 15 to 17. No new scan was performed.

Test point	Findings
Date	16.10.2026, approx. 08:32
Source	192.168.20.7 - Kali
Objective	192.168.20.20 - Ubuntu
Network target	TCP port 199
Suricata signature	SOC LAB Possible TCP SYN Port Scan
Suricataity assessment	Severity 2; Action: allowed
Wazuh detection	Rule ID 100121, level 8
MITRE ATT&CK	T1046 - Network Service Discovery
Evaluation	True Positives - Authorized Network Scan
Reaction	No containment required; analytically closed

Analyst rating Network activity was actually generated and correctly recognized by Suricata as a possible TCP-SYN port scan. The source host 192.168.20.7 is the well-known potash system of the isolated laboratory. The target and timing are consistent with the authorized detection test. Thus, the alert is a true positive, but not a security incident.

Sensor effect The value action=allowed confirms IDS operation: Suricata detected and logged traffic but did not block it. This behavior corresponds to the intended monitoring function of the laboratory.

Closure Note Authorized TCP SYN scan from Kali 192.168.20.7 to Ubuntu 192.168.20.20 confirmed. Suricata and Wazuh Rule 100121 detected the activity as designed. No unauthorized source, follow-up compromise or containment requirement identified.

19.4 Triage of a Domain Admins Change

As the third Triage case, a privileged group change was investigated on the domain controller DC-Test. The existing alert to Rule 60159 was evaluated over a period of three months; a total of four hits were available. The latest event was analysed. No new change was carried out in Active Directory.

Test point	Findings
Date	17.09.2026, 09:52:38 a.m.
System	DC-Test - Domain Controller
Windows event	Event ID 4729 - member removed from security group
Account concerned	SOC Rule Test
Privileged Group	Domain Admins
Executing account	Administrator
Wazuh detection	Rule ID 60159, level 12
MITRE ATT&CK	T1098.007 - Additional Local or Domain Groups
Tactics	Persistence; Privilege Escalation
Evaluation	True Positives - Authorized Administrative Change
Reaction	No containment required; analytically closed

Analyst rating The removal of the account SOC Rule Test from the group Domain Admins was actually logged and correctly recognized as a high-severity alert by Wazuh. The running account administrator and purpose and time agree with the authorized laboratory test. Therefore, the event is a true positive, but does not constitute a security incident.

Unauthorized Change Response In a production environment, group membership would be immediately checked and reset to the released state. Additionally, the administrator account, other privileged changes and associated login events would have to be examined.

Closure Note Authorized removal of SOC Rule Test from Domain Admins confirmed on DC-Test. Windows Event ID 4729 and Wazuh Rule 60159 (Level 12) detected the change as designed. No unauthorized privilege modification or containment requirement identified.

19.5 Uniform SOC-Triage-Runbook

This runbook standardizes the processing of Wazuh alerts in the Blue Team Lab. It applies to endpoint, network and active directory events and ensures that each alert is validated, evaluated, treated and closed in a comprehensible manner.

Standard sequence

No.	Phase	Test steps and documentation
1	Record alert	Date, Agent/System, Rule ID, Alert level and description.
2	Technical validation	Check raw event: Event ID, signature, command, user, source, destination, port and protocol - depending on the alert type.
3	Check context	Match with test plan, change window and responsible person. Do not suspect authorization, but prove it.
4	Determine the scope	Search for contiguous alerts, other hosts or accounts, repetitions and time proximity.
5	Evaluate risk	MITRE technology and tactics, criticality of the asset, Wazuh level and possible impacts.
6	Classify	True Positives - Authorized Activity, True Positives - Security Incident, False Positives or Inclusive / Escalation Required.
7	React	Document and close in case of authorized activity. Secure, contain evidence in case of an incident, inform the competent authority and initiate further analysis.
8	Complete	Completely document the result, justification, measures, analyst, time stamp and closure note.

Escalation criteria

- Alert level 12 or higher or a particularly critical system is affected.
- Privileged accounts or groups such as Domain Admins have been changed.
- A coded or otherwise conspicuous command is not clearly authorized.
- The source is unknown, the activity repeats or affects several systems.
- There are indications of successful execution, persistence, rights expansion or data outflow.
- The authorization cannot be reliably proven.

Application to documented use cases

Use Case	Detection	Classification	Decision
PowerShell	Rule 92057 / Level 12	True Positives - Authorized Activity	No containment; documented and closed
Suricata portscan	Rule 100121 / Level 8	True Positives - Authorized Activity	No containment; documented and closed
Domain Admins	Rule 60159 / Level 12	True Positives - Authorized Activity	No containment; documented and closed

Alert type reaction guideline

Sector	Primary test	Possible reaction to unauthorized activity
Endpoint	Process tree, user, command line, Sysmon Event ID	Isolate host, investigate process, secure evidence and check account.
Network	Source, destination, port, protocol, frequency and other objectives	Block or segment the source, check affected targets and secure traffic.
Active Directory	Event ID, executive account, target account and group change	Reset Privileged Change, Lock or Reset Account and check for further AD changes.

Reusable Closure Note

Template Alert [Rule ID] to [System] was validated on the basis of [Events/Beweis]. The activity was assessed as [classification] because [reason]. Measures implemented: [measures or no containment required]. The case was closed or escalated at [Analyst] at [Analyst].

Result The Runbook combines the three successfully analyzed use cases with a uniform, reusable SOC process. Thus, triage decisions in the project are consistent, verifiable and documented for a portfolio.

19.6 Incident response run at PowerShell

The following procedure transfers the uniform triage runbook to the already validated PowerShell alert. The event was created by an authorized laboratory test and was not a real security incident. The incident response measures are therefore documented as a controlled tablet scenario; no productive accounts were blocked and no system isolated.

Situation

Test point	Findings
System concerned	ham1 - Windows 11 Endpoint
Users	TEST\Administrator
Detection	Wazuh Rule 92057, level 12; Sysmon Event ID 1
Observed Command	PowerShell with EncodedCommand; decoded content: Write output "SOC test"
MITRE ATT&CK	T1059.001 - PowerShell; Tactics: Execution
Triage Result	True Positives - Authorized Activity
Incident status	No real incident; Tabletop response documented

Incident Response Phases

Phase	Procedure in the scenario	Result/evidence
1. Preparation	Wazuh agent, Sysmon process monitoring and the integrated detection rule are activated. Responsibility, evidence sources and triage runbook are defined.	Endpoint telemetry and processing process available.
2. Detection & Analysis	Alert, Sysmon Event ID 1, User and Command Line. Decode EncodedCommand, match with the test plan and search for follow-up events.	Authorized test confirmed; no evidence of follow-up activity.
3. Containment	In case of unauthorized activity: logically isolate endpoint, stop suspicious process and limit account activity. Do not perform containment in the confirmed laboratory test.	Containment not required; decision documented.
Employment and social policy	Confirmed attack: remove malicious scripts, tasks, or persistence, identify cause, and reset compromised access data.	No harmful acts or persistence detected.
5. Recovery	Check system status, Wazuh-Agent and Sysmon-Telemetry. If the endpoint is real isolation, only release and monitor the endpoint after validation.	System remained operational; telemetry worked.
Lessons Learned	Detection, alert context and reaction decision evaluate. In future, authorized tests will clearly identify and use Closure Note standardized.	Detection works; Documentation and Runbook reusable.

Decision-making and escalation logic

Condition	Decision	Measure
Test clearly authorized; no follow-up activity	Authorized True Positives	Document and close analytically
Authorization unclear or command unknown	Inclusive	escalate to those responsible; secure evidence
Unknown source, harmful content or subsequent activity	Security Incident	Isolate Endpoint; examine account, process tree and other hosts
Detection technically incorrect or expected normal behaviour	False positives	Check the rule context and perform tuning only after proof

Evidence to be secured

- Wazuh alert with Rule ID, Level, Agent, Time stamp and full event context.
- Sysmon Event ID 1 with process, parent process, user and complete command line.
- Decoding PowerShell content and verifiable comparison with the authorized test plan.
- Related alerts and process events in the relevant time window.
- Analyst decision, measures, responsible person and closing date.

Communication and completion

Communication In case of a real incident, Incident Owners and System Managers are informed. Status, risk, affected assets and measures already carried out are recorded in a concise and verifiable manner.

Closure Note: Wazuh Rule 92057 (Level 12) detected an encoded PowerShell command on ham1 through Sysmon Event ID 1. The command decoded to Write-Output "SOC-Test" and matched the authorized laboratory test. No malicious follow-up activity, persistence, or containment requirement was identified. The alert was classified as True Positive – Authorized Activity and closed analytically.

Result The Use Case shows not only the technical recognition, but also a complete and comprehensible incident response decision-making process. Real and only simulated measures are clearly separated; this keeps the project documentation technically correct.

20. MITRE ATT&CK and Event Types

This reference explains the terms used in the project. It describes categories and identifiers; it is not an additional test procedure.

MITRE ATT&CK Structure

Term	Meaning
Tactic	The adversary's high-level objective, such as Execution, Discovery, Persistence, or Privilege Escalation.
Technique	Describes how an objective may be achieved; ID format Txxxx, for example T1046.
Sub-technique	A more specific variant of a Technique; ID format Txxxx.xxx, for example T1059.001.
Procedure	A concrete observed implementation of a Technique by a group, software, or activity.
Mapping	Analytical assignment of observed behavior. It does not replace technical Event validation.

Event and Identifier Types

Type	Example in this project	Meaning
Windows Event ID	4625, 4728, 4729	Event type logged by Windows in the applicable Event Log.
Sysmon Event ID	1 – Process Create	Endpoint Telemetry generated by Sysmon; Event ID 1 records process creation.
PowerShell Event ID	4104 – Script Block Logging	PowerShell Log Event containing captured Script Block content when Logging is enabled.
Suricata SID	1000001, 1000002	Unique Signature ID for a Suricata Rule. It is not a Wazuh Rule ID.
Wazuh Rule ID	92057, 60159, 100121	Identifier of the Wazuh Rule that evaluates a decoded Event and may generate an Alert.
Alert Level / Severity	Wazuh Level 8/12; Suricata Severity 2	Product-specific prioritization. Values from different products are not directly equivalent.
Raw Event / Alert	Windows/Sysmon/eve.json → Wazuh Alert	A Raw Event is Telemetry; an Alert is generated only when a Detection Rule matches.

Project mappings: T1059.001 = PowerShell (Execution); T1046 = Network Service Discovery (Discovery); T1098.007 = Additional Local or Domain Groups (Persistence, Privilege Escalation).